

**INTELLIGENT CO-ORDINATED EV CHARGING
STRATEGY FOR VOLTAGE AND THERMAL
STABILITY IN WEAK DISTRIBUTION GRIDS WITH
HIGH PV PENETRATION**

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Sri Lanka

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Thesis submitted in partial fulfillment of the requirements for the Honours Degree of
Bachelor of Science in Engineering

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DECLARATION

We declare that this is our own work and this report does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

The rapid growth of rooftop solar photovoltaic (PV) generation and electric-vehicle (EV) charging is placing weak distribution feeders under stress they were never designed to absorb. High PV penetration raises midday voltages and introduces fluctuation, while clustered evening EV charging causes undervoltage and thermal overloading of transformers and feeders. This thesis develops an intelligent coordinated EV charging strategy to maintain voltage stability and thermal security on such networks while still meeting vehicle charging demand.

A two layer hierarchical controller is proposed. An upper-layer particle swarm optimization (PSO) determines the number of active chargers at each aggregation in every 15-minute interval, evaluating candidates against the true network state through a backward/forward sweep load flow with a network-aware objective covering voltage violation, PV induced fluctuation, energy backlog, and transformer overload. A lower-layer Mamdani fuzzy controller then allocates the permitted chargers to individual vehicles by a priority score combining energy demand, remaining time, and fairness. The strategy is tested in MATLAB on a deliberately weakened IEEE 33 bus, 11 kV feeder with five PV sites and five EV aggregations, over a full day, against an uncoordinated baseline.

Whereas the uncoordinated case breached the 0.94 p.u. voltage limit, overloaded the 10 MVA transformer and the main feeder, the coordinated strategy held all voltages within limits, shaved the transformer peak to its rating, reduced night-peak feeder loading to 92.32%, cut network losses by 11.6% and charging cost by 11.7%, and improved the voltage-stability margin at every bus, while satisfying 99.49% of energy demand. Coordinated PSO-fuzzy control can thus stabilize weak, high PV feeders under heavy EV penetration using charging flexibility alone. Future work includes sensitivity and Monte Carlo analysis, vehicle-to-grid support, and real-time deployment.

Keywords: Coordinated EV charging; Particle swarm optimization; Fuzzy logic control; Voltage and thermal stability; Weak distribution network with high PV penetration

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B. Contribution to the Design Component.

I was personally responsible for the core control strategy. I designed and implemented the load flow model, the particle swarm optimization layer and its network aware cost function and built the Mamdani fuzzy controller the inference system, assigning membership functions, rule base, and the priority-based charger allocation. I integrated these layers with the load flow in the main coordinated simulation and generated the result visualizations. I also designed the dashboard for project demonstration.

C. Contribution to the Report.

I authored Chapter 3 (Methodology), Chapter 4 (Simulation Setup), and Chapter 5 (Results and Discussion), wrote the Abstract, and compiled the consolidated reference list. I produced the objective function and fuzzy-system formulations and the comparative results tables.



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B. Contribution to the Design Component.

I implemented the uncoordinated baseline simulation used as the comparison case, and the voltage stability index computation applied to every bus. I also assisted with the performance-metric calculations and the plotting of results, including the EV charging-status visualizations. I also implemented some of the visualizations related to fuzzy controller and the project dashboard for demonstration.

C. Contribution to the Report.

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B. Contribution to the Design Component.

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C. Contribution to the Report.

I authored Chapter 6 (Conclusion and Future Work) and rest of the documentation, assisted with the report formatting and the lists of figures and tables, and contributed to the project dashboard.



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List of Abbreviations

Abbreviation	Definition
AAAC	All Aluminium Alloy Conductor
AC	Alternating Current
BFS	Backward/Forward Sweep (load flow)
CEB	Ceylon Electricity Board
DC	Direct Current
DG	Distributed Generation
EV	Electric Vehicle
FIS	Fuzzy Inference System
FLC	Fuzzy Logic Controller
GA	Genetic Algorithm
IEEE	Institute of Electrical and Electronics Engineers
kV	Kilovolt
kW	Kilowatt
kWh	Kilowatt-hour
LECO	Lanka Electricity Company (Pvt) Ltd
LKR	Sri Lankan Rupee
LV	Low Voltage
MILP	Mixed-Integer Linear Programming
MV	Medium Voltage
MVA	Megavolt-ampere
MW	Megawatt
MWp	Megawatt-peak
OLTC	On-Load Tap Changer
PSO	Particle Swarm Optimization
PUCSL	Public Utilities Commission of Sri Lanka
PV	Photovoltaic
p.f.	Power Factor
p.u.	Per Unit
SOC	State of Charge
ToU	Time-of-Use
V2G	Vehicle-to-Grid
VSI	Voltage Stability Index

List of Symbols

Symbol	Description	Unit
a, b, c	Membership-function shape parameters	–
C_{bat}	EV battery capacity	kWh
C	Total charging cost	LKR
c_1, c_2	PSO cognitive and social coefficients	–
ed, rt, fi	Fuzzy inputs: energy demand, remaining time, fairness index	–, intervals
E_{req}	Required (grid-side) charging energy of a vehicle	kWh
E_{rem}	Remaining charging energy of a vehicle	kWh
$E_{\text{rem}}^{\text{tot}}$	Total remaining energy backlog	kWh
E_{loss}	Total daily network energy loss	kWh
g, p	PSO global-best and personal-best positions	–
$h(t), p(t)$	Normalized domestic load and PV profiles	p.u.
I_i	Bus current	A (p.u.)
I_1	Main-feeder current	A
I_{amp}	Conductor ampacity	A
J	Objective (cost) function value	–
$J_1\text{--}J_4$	Objective terms: voltage, fluctuation, backlog, thermal	–
J_b	Branch current (backward sweep)	p.u.
L_{np}	Average night-peak feeder loading	%
N_s	Number of active chargers at station s	–
$N_{\text{avail},s}$	Number of available vehicles at station s	–
N_{opt}	Optimal charger-count vector (PSO output)	–
P, Q	Active and reactive power	kW, kVAr
P_{ch}	Charger power rating	kW
P_{pv}	PV unit rating	kW
P_{tot}	Total charging power	kW
P_i (priority)	Charging priority score of vehicle i (fuzzy output)	– (0–1)
R, X	Branch resistance and reactance	Ω (p.u.)
r_1, r_2	Random numbers in [0, 1]	–
$S_{\text{base}}, V_{\text{base}}, Z_{\text{base}}$	System base power, voltage, impedance	MVA, kV, Ω
$S_{\text{load},i}$	Complex domestic load at bus i	p.u.
$S_{\text{net},i}$	Net complex power injection at bus i	p.u.
$S_{\text{pv},k}$	PV generation at bus k	p.u.
S_1	Transformer (slack-bus) apparent power	MVA

S_T	Transformer rating	MVA
$S_{T,peak}$	Transformer peak loading	MVA
SOC	State of charge	%
SOC_{arr}, SOC_{exp}	Arrival and target state of charge	%
t	Time-interval index (1–96)	–
t_{arr}, t_{dep}	Arrival and departure interval of a vehicle	–
V_i	Bus voltage magnitude	p.u.
V_{min}, V_{max}	Lower and upper voltage limits	p.u.
V_s, V_r	Sending- and receiving-end branch voltages	p.u.
VSI	Voltage stability index	–
v	PSO particle velocity	–
w	PSO inertia weight	–
w_1-w_4	Objective-function weights	–
x	PSO particle position (charger-count vector)	–
z_{pu}, z_{Ω}	Per-unit and ohmic branch impedance	p.u., Ω
Greek symbols		
Δt	Time-step length (15 min)	h
ΔS_{peak}	Transformer peak shaving	MVA
ε	Load-flow convergence tolerance	p.u.
η_{ch}	Charging efficiency	–
η_{sat}	Energy-demand satisfaction	%
κ	Domestic load scaling factor	–
$\mu(z)$	Aggregated fuzzy output membership	–
$\pi(t)$	Time-of-use electricity tariff	LKR/kWh
Ω_s	Set of vehicles assigned to station s	–

Chapter 1

1. INTRODUCTION

1.1. Background

Sri Lanka's electricity distribution network is undergoing rapid change due to the increasing adoption of rooftop photovoltaic (PV) systems and electric vehicles (EVs). Government initiatives such as net metering, net accounting, net plus, and the Soorya Bala Sangramaya program have accelerated rooftop PV installations, while declining battery costs and supportive import policies have encouraged EV adoption. Although these technologies contribute significantly to national decarbonization goals, they also introduce new operational challenges. Existing distribution networks, originally designed for one-way power flow and predictable residential demand, are now required to accommodate bidirectional power flow, variable renewable generation, and growing EV charging demand, creating voltage and thermal management issues [1], [2].

Most medium voltage distribution networks operated by CEB and LECO are based on 11 kV radial feeders. Although radial networks are simple, economical, and easy to protect, they provide limited voltage support because power is supplied through a single path from the substation. Consequently, voltage decreases along the feeder as loading increases, making long or heavily loaded feeders particularly vulnerable to voltage instability. Such systems are referred to as electrically weak, as their voltages are highly sensitive to changes in power flow. To reflect these conditions, this study employs a modified IEEE 33-bus radial distribution network configured as a weak 11 kV feeder, enabling the impacts of PV generation and EV charging to be clearly evaluated.

The PV generation and EV charging affect distribution networks at different times of the day, creating contrasting operational challenges. During midday, high PV generation can reverse power flow, increase local voltages, and introduce rapid voltage fluctuations due to cloud-induced intermittency [1]. In contrast, EV charging demand is typically concentrated during the evening when most vehicles return home, coinciding with peak residential demand. This causes voltage drops at remote buses and increases transformer and feeder loading, potentially leading to thermal overloading

[3]. Consequently, the same distribution feeder may experience overvoltage during the day and undervoltage with equipment overloading in the evening, making coordinated management essential.

In uncoordinated charging, EVs begin charging immediately upon connection at full power, creating a synchronized increase in demand during the evening peak. This unmanaged behavior intensifies network loading and serves as the baseline for evaluating the effectiveness of the proposed coordinated charging strategy.

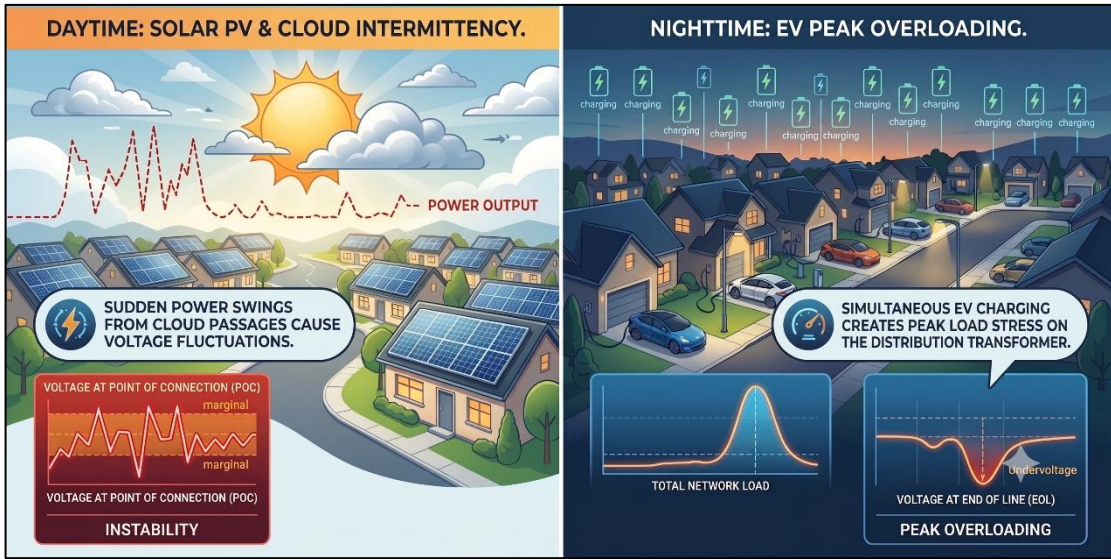


Figure 1 - Issues of PV and EV in distribution networks

1.2. Motivation

Unlike conventional household appliances, EV charging demand is flexible because vehicles only need to reach their required state of charge before departure rather than being charged immediately. Since EVs are typically parked for longer than the time needed to recharge, their charging can be shifted within this available time window. This flexibility enables charging demand to be redistributed across the day, reducing peak loading while still meeting users' energy requirements, making EVs valuable as controllable loads in modern distribution networks.

Managing large numbers of EVs individually is impractical; therefore, vehicles are grouped into aggregations that can be controlled as single charging units by the network operator [4]. Coordinated charging at the aggregator level helps maintain acceptable

voltage levels, reduces transformer and feeder loading, and shifts charging demand away from the evening peak to periods with available network capacity and higher PV generation [5]. For utilities such as CEB, this approach provides a cost-effective alternative to expensive network reinforcement by making better use of existing infrastructure. Additionally, coordinated charging increases the utilization of daytime PV generation, reducing renewable energy curtailment and improving overall network efficiency.

EV charging control must adapt to continuously changing network conditions caused by variable PV generation and stochastic vehicle behavior. To address this, the proposed two-layer framework combines PSO for network-level charging optimization with a fuzzy logic controller for fast, real-time allocation of chargers to individual vehicles while maintaining voltage and thermal limits.

1.3. Problem Statement

A weak 11 kV radial distribution feeder with high PV penetration and multiple EV charging aggregations faces significant operational challenges under uncoordinated charging. During the evening, simultaneous EV charging causes voltage drops at remote buses, transformer overloading, and feeder currents approaching or exceeding conductor limits [6]. Conversely, high midday PV generation produces voltage rise and fluctuations due to intermittent cloud cover. As a result, the feeder experiences both voltage stability and thermal security issues at different times of the day, despite the overall daily EV energy demand remaining relatively modest.

This research addresses the problem of determining, at every 15-minute interval, the number of chargers to operate at each EV aggregation and the vehicles to be served. The objective is to maintain voltage and thermal limits while maximizing the charging energy delivered before vehicle departure. Since charging decisions are interconnected over time and influenced by uncertain PV generation and EV behavior, a coordinated optimization and fuzzy inference framework is required instead of simple rule-based charging.

1.4. Research Objectives

1.4.1. Main Objective

To develop and evaluate an intelligent coordinated EV charging strategy that maintains voltage stability and thermal security in a weak distribution grid with high PV penetration, while satisfying the charging requirements of the connected electric vehicles.

1.4.2. Specific Objectives

- Model a weak 11 kV modified IEEE 33-bus radial distribution network with high PV penetration and multiple EV charging aggregations using backward–forward sweep load flow.
- Establish an uncoordinated charging baseline to quantify voltage violations and thermal overloading.
- Develop a PSO-based optimization layer to determine the optimal number of active chargers while considering voltage and thermal constraints.
- Design a fuzzy logic controller to allocate chargers based on energy demand, remaining parking time, and fairness.
- Evaluate the proposed coordinated strategy using voltage profile, transformer and feeder loading, peak shaving, charging satisfaction, and network loss reduction.

1.5. Scope of the Study

This study is carried out entirely through MATLAB simulations over a representative 24-hour period divided into 96 fifteen-minute intervals. The test system is a modified IEEE 33-bus radial distribution network operating under weak-grid conditions with five PV stations and five EV charging aggregations. Network performance is evaluated using a backward–forward sweep load flow. Daily load and PV generation profiles are applied, with PV intermittency represented through added fluctuations, while EV characteristics, including arrival times, departure times, and charging demands, are generated stochastically to reflect realistic charging behavior.

This research focuses solely on coordinated EV charging by controlling the timing and number of active chargers. Vehicle-to-grid operation, reactive power support, tap-changing transformers, capacitor switching, and network reconfiguration are not considered. The study assumes a constant charging efficiency of 90% and evaluates performance using a single modified IEEE 33-bus feeder over one representative day. Communication infrastructure and practical implementation aspects are discussed conceptually, while comprehensive sensitivity analysis and Monte Carlo simulations are identified as future research directions.

1.6. Research Contributions

- Developed a two-layer coordinated charging framework that separates network-level charger scheduling from vehicle-level charger allocation.
- Proposed a network-aware PSO optimization method with embedded load flow analysis to accurately evaluate voltage and thermal constraints.
- Designed a fuzzy logic controller that prioritizes EV charging using energy demand, remaining parking time, and fairness.
- Validated the proposed framework on a weak, high-PV distribution feeder, demonstrating improved voltage stability, reduced thermal loading, effective peak shaving, and high charging satisfaction with minimal energy compromise.

1.7. Organization of the Report

The remainder of this thesis is organized into seven chapters. Chapter 2 reviews the literature on PV integration, EV charging, aggregation, optimization techniques, intelligent control methods, and voltage and thermal stability issues, highlighting the research gap addressed in this work. Chapter 3 presents the methodology starting with system model and problem formulation and then describes the proposed coordinated EV charging strategy, including the PSO based charger scheduling and fuzzy logic allocation framework. Chapter 4 explains the simulation setup, datasets, scenarios, and evaluation metrics. Chapter 5 presents the simulation results and compares coordinated and uncoordinated charging performance. Finally, Chapter 6 summarizes the key findings, outlines the main contributions, and suggests directions for future research.

Chapter 2

2. LITERATURE REVIEW

This chapter reviews the literature on coordinated EV charging in weak distribution networks with high PV penetration. It examines the impacts of distributed PV generation, EV charging demand, and charging infrastructure on distribution systems. The chapter then discusses EV aggregation strategies, optimization methods, and intelligent control techniques used for coordinated charging. Finally, it reviews voltage and thermal stability challenges to identify existing research gaps. This review establishes the motivation for the proposed coordinated charging framework and highlights how the present work addresses the limitations of existing approaches.

2.1. Distribution Networks with High PV Penetration

Distributed rooftop PV generation fundamentally changes the operation of distribution networks by introducing generation at the consumer level. Unlike traditional feeders designed for one-way power flow, high PV output can cause reverse power flow when local generation exceeds demand, leading to voltage rise at the point of connection. This effect is particularly significant in weak distribution feeders with high line impedance, where voltage becomes highly sensitive to power injections. Maghami et al. demonstrated that increasing PV penetration progressively degrades voltage regulation and power quality, with the impacts becoming more severe as PV capacity approaches the feeder's operating limits [1].

A second difficulty is that PV output is not steady. Generation rises and falls with the sun over the day, but it also varies on a timescale of seconds to minutes as cloud passes over the panels. The resulting swings in injected power appear at the connection point as voltage fluctuation, and in the worst case as flicker. For a controller concerned with voltage quality, it is this short-term variability, concentrated around the midday generation peak, that is hardest to manage, because it cannot be predicted accurately in advance.

Conventional voltage regulation methods, including on-load tap changers, capacitor banks, and reactive power compensation, are widely used to maintain acceptable

voltage levels in distribution networks. In addition, optimal PV placement and sizing can reduce the adverse impacts of distributed generation. For example, Rajakumar et al. applied a metaheuristic optimization technique to determine optimal PV locations and capacities for improved network performance [7]. Although these solutions are effective, they require significant investment and infrastructure upgrades while treating demand as uncontrollable. In contrast, coordinated EV charging utilizes the inherent flexibility of charging demand, offering a cost-effective alternative that is particularly attractive for rapidly evolving distribution networks such as those in Sri Lanka.

2.2. Electric Vehicle Charging Infrastructure

EV charging infrastructure ranges from low-power residential chargers to high-power DC fast chargers. This study focuses on 11 kW AC chargers commonly used in residential and workplace charging stations. Since vehicles remain connected for relatively long periods, their charging can be shifted within the available parking duration without affecting user requirements. This charging flexibility makes AC charging particularly suitable for coordinated scheduling, unlike fast DC charging, where short connection times provide limited opportunity for demand management.

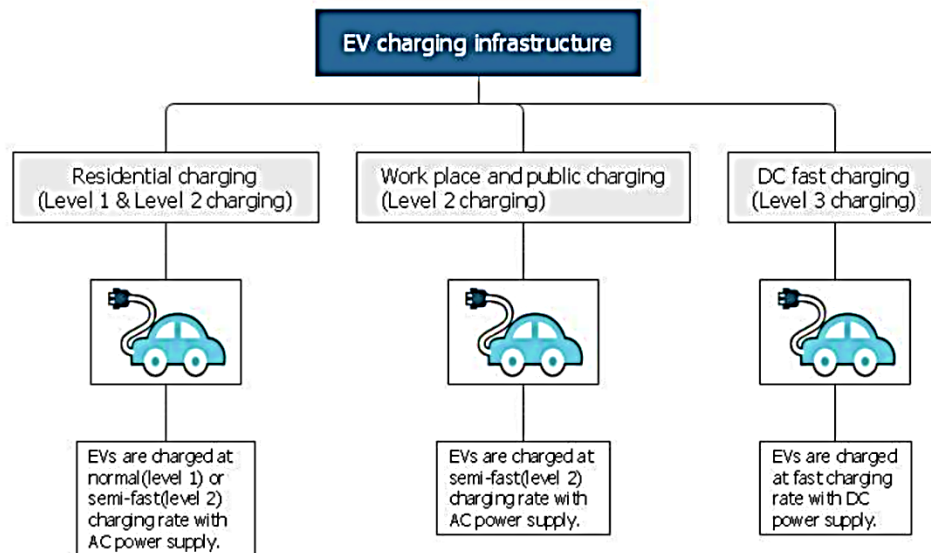


Figure 2 - EV charger levels and use cases

Extensive research has focused on the planning of EV charging infrastructure, particularly the optimal placement and sizing of charging stations. Ferraz et al. reviewed this field and found that most studies formulate charging station planning as

an optimization problem that considers both technical and economic constraints [5]. However, comparatively fewer studies address the operational management of existing charging stations. Specifically, limited attention has been given to determining how charging should be coordinated in real time to maintain network voltage and thermal limits once the charging infrastructure has already been deployed.

Effective EV charging coordination requires realistic modelling of vehicle behavior, as arrival times, departure times, initial state of charge, and energy demand vary among users and across days. Recent charging transaction datasets, such as the one published by Baek et al., provide valuable information for representing these variations [8]. In this study, stochastic EV profiles are generated to reflect realistic charging behavior by incorporating diverse arrival and departure patterns together with varying charging energy requirements, ensuring that the simulation closely represents real-world EV charging conditions.

2.3. Impact of EV Charging on Distribution Networks

The impact of EVs on distribution networks depends largely on when charging occurs. In residential areas, most vehicles begin charging during the evening after users return home, causing EV demand to coincide with the existing residential peak. This synchronized charging significantly increases network loading, resulting in greater voltage drops, equipment stress, and higher power losses.

Numerous studies have shown that uncoordinated EV charging can cause significant operational problems, including voltage drops, transformer overloading, increased feeder loading, and higher network losses, particularly in radial distribution systems. Nandi et al. demonstrated that electrically weak buses are the first to experience voltage violations when EV charging demand increases [6]. Likewise, Ferraz et al. identified voltage regulation and thermal loading as major challenges in the planning and operation of EV charging infrastructure, especially in distribution networks with limited capacity [5].

These findings motivate the present study. Uncoordinated charging is used as the baseline for performance evaluation, while the proposed coordinated strategy is

designed to simultaneously manage voltage regulation and thermal loading in weak distribution networks with high PV penetration throughout the entire day.

2.4. EV Aggregation Strategies

As the number of EVs connected to the grid continues to increase, controlling each vehicle individually becomes computationally challenging. To address this, most coordinated charging strategies use an aggregator, which groups multiple EVs connected at a common location and represents them as a single controllable load. This simplifies the control problem while providing an effective interface between the distribution network operator and individual vehicles.

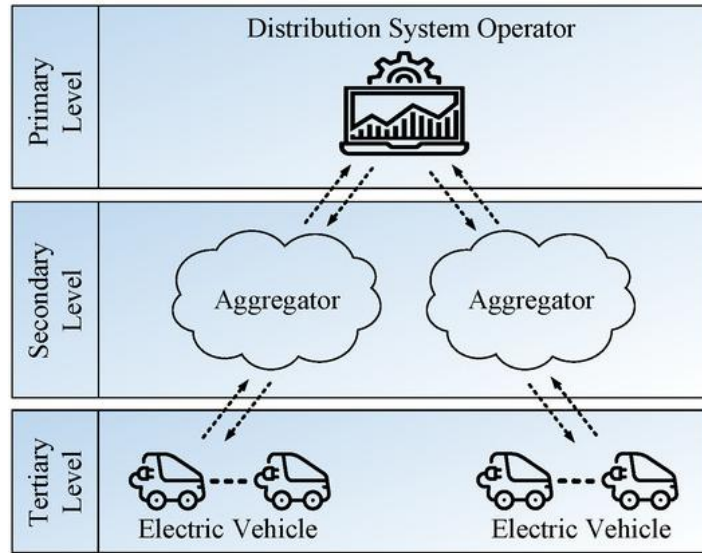


Figure 3 - Connection between DSO, Aggregator and EV users

Aggregator-based charging has been implemented using centralized, decentralized, and hierarchical control architectures. For example, Ghatak et al. proposed a collaborative scheduling framework that integrates vehicle-to-grid capability with renewable energy resources, demonstrating the potential of aggregated EVs to support both charging requirements and grid operation [9]. However, many existing studies model the aggregated EV demand as a continuous load and do not explicitly determine which vehicles should be charged at each interval or consider fairness among users. The proposed approach addresses this limitation by combining aggregator-level scheduling with a vehicle-level fuzzy controller, enabling both efficient grid management and fair charger allocation.

2.5. Optimization Techniques for EV Charging

Coordinated EV charging is commonly formulated as a constrained optimization problem that minimizes objectives such as voltage deviation, power loss, or unmet charging demand while satisfying network and charging constraints. Since the problem is nonlinear due to load flow interactions, the optimization method must balance solution quality with computational efficiency. Several approaches have been proposed in literature.

2.5.1. Particle Swarm Optimization (PSO)

PSO, introduced by Kennedy and Eberhart [10], is a population-based optimization technique inspired by the collective behavior of bird flocks and fish schools. In PSO, multiple candidate solutions, known as particles, explore the search space by updating their positions based on their own experience and the best solution found by the swarm. Since PSO does not require gradient information, it can be directly integrated with nonlinear load flow calculations, making it well suited for power system optimization problems. It is also capable of handling non-convex objective functions with relatively few tuning parameters and fast convergence. Although PSO cannot guarantee the global optimum and its performance depends on appropriate parameter selection, it has been widely adopted for distribution network applications. Improved PSO variants have also demonstrated excellent performance in multi objective EV charging and discharging scheduling, confirming their suitability for coordinated charging problems in modern distribution systems [11].

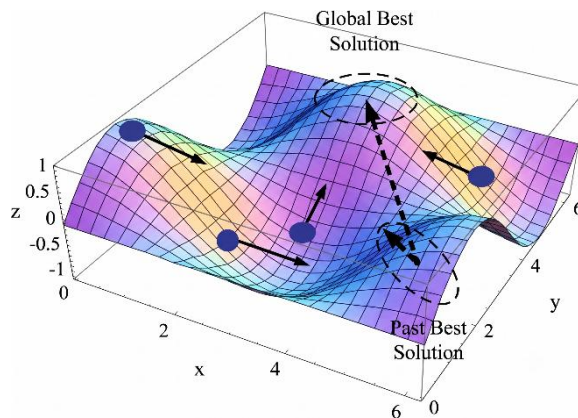


Figure 4 - PSO algorithm optimum solution finding logic

2.5.2. Genetic Algorithm (GA)

Genetic Algorithms (GAs), introduced by Holland [12], are evolutionary optimization methods suitable for nonlinear and combinatorial problems. Although widely used in power system scheduling, they generally require more function evaluations and parameter tuning than PSO, making them less computationally efficient for load flow-based coordinated EV charging optimization.

2.5.3. Mixed-Integer Linear Programming (MILP)

Mixed-integer Linear Programming (MILP) provides optimal solutions for problems with discrete decision variables and linear constraints. However, EV charging coordination requires nonlinear AC load flow modelling, which must be simplified for MILP. As problem size increases, computational complexity also grows, limiting its suitability for repeated real-time network optimization.

2.5.4. Other Optimization Methods

Several other optimization techniques, including Artificial Bee Colony, Grey Wolf Optimizer, and Jellyfish Search Optimization, have also been applied to EV and PV integration problems. For example, Rajakumar et al. used Jellyfish Search Optimization for optimal PV siting and sizing [7]. Convex relaxation methods offer efficient alternatives but involve trade-offs between computational efficiency, solution quality, and constraint handling.

Technique	Nature	Key strengths	Key limitations
PSO	Population-based metaheuristic	Gradient-free; embeds nonlinear load flow directly; few parameters; fast convergence	No global-optimum guarantee; performance sensitive to tuning
GA	Evolutionary metaheuristic	Robust global search; handles combinatorial and non-convex problems	Slower convergence; more operators and parameters to tune; heavier per run
MILP	Exact mathematical programming	Provably optimal; clean handling of discrete decisions and linear constraints	Requires linearized load flow; scales poorly with problem size; less suited to real-time AC networks

Technique	Nature	Key strengths	Key limitations
Other metaheuristics / convex relaxation	Heuristic or reformulation-based	Flexible; some offer efficient, near-optimal solutions	Varied reliability; relaxations may not hold under all conditions

Table 1: Comparison of optimization techniques commonly applied to our problems

Most optimization based EV charging methods are centralized and solve static scheduling problems using aggregated, continuous loads. Comparatively few studies address real time adaptation or individual vehicle allocation. This limitation motivates the proposed two layer framework, where optimization performs grid level scheduling and a fast controller manages vehicle-level charging decisions.

2.6. Intelligent Control Techniques

Where optimization answers the question of how much charging the network can support, a second question remains: how should limited charging capacity be distributed among competing vehicles, quickly and repeatedly, as conditions change. This is a control problem rather than a planning problem, and intelligent-control techniques are well suited to it.

2.6.1. Fuzzy Logic

Fuzzy logic, introduced by Zadeh through fuzzy set theory [13] and later applied to control systems by Mamdani and Assilian [14], provides an effective way to handle uncertainty using linguistic rules instead of precise mathematical models. Unlike conventional rule-based controllers with fixed thresholds, fuzzy logic produces smooth control actions and can accommodate imprecise or varying inputs. It is computationally efficient, easy to interpret, and well suited for real-time decision-making. These advantages have led to its application in coordinated EV charging for LV distribution networks [15] and in prioritizing vehicles competing for limited charging resources [16]. Owing to its simplicity and fast execution, fuzzy logic is particularly suitable for vehicle-level charging decisions. In this research, a Mamdani fuzzy inference system is employed to prioritize EVs by considering three key factors: energy demand, remaining parking time, and waiting fairness, enabling efficient and equitable charger allocation in real time.

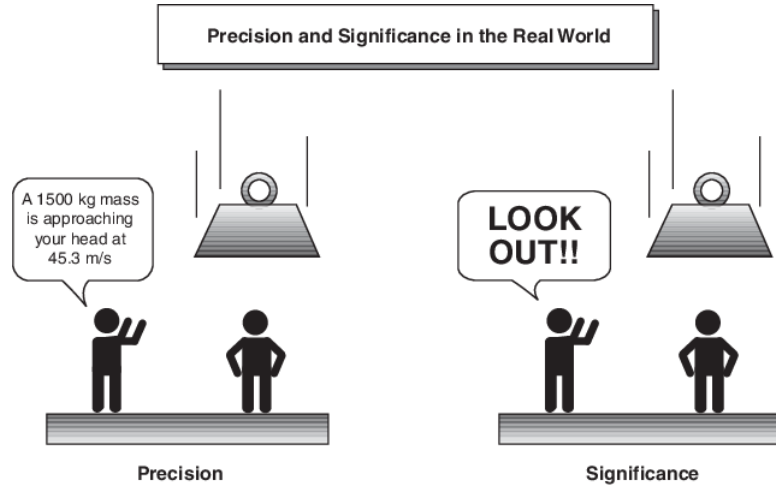


Figure 5 - Difference between precision and significance

2.6.2. Other AI-Based Methods

Other AI techniques such as artificial neural networks, neuro-fuzzy systems, and reinforcement learning have also been explored for EV charging control. While these methods can learn complex decision patterns, they typically require large training datasets and often lack interpretability. Moreover, ensuring strict compliance with network constraints is more challenging. Therefore, this work adopts a transparent PSO-based optimization framework with a fuzzy logic controller, reserving learning-based methods as potential future extensions.

2.7. Voltage and Thermal Stability in Distribution Networks

Voltage stability refers to the ability of a distribution network to maintain acceptable voltage levels at all buses under varying loading and generation conditions. In radial distribution systems, increasing demand causes voltage to decrease along the feeder, and excessive loading can eventually lead to voltage instability. To assess the proximity of a bus to this condition, voltage stability indices (VSIs) are widely used. These indices combine network parameters and power flow information into a single value that indicates the available stability margin [17].

Nandi et al. applied a VSI to evaluate the impact of fast EV charging stations on a radial distribution network, identifying electrically weak buses that are more susceptible to voltage violations under increased loading [6]. In this research, the same approach is

adopted to evaluate voltage stability throughout the network and compare the effectiveness of coordinated and uncoordinated charging strategies.

Thermal loading is a key operational constraint because excessive transformer and feeder loading accelerates equipment ageing and increases failure risk. Unlike many previous studies, this research simultaneously addresses voltage stability, thermal constraints, and PV-induced voltage fluctuations, providing a more comprehensive coordinated EV charging strategy for weak distribution networks.

2.8. Research Gaps

1. Existing studies rarely investigate weak distribution feeders with both high PV penetration and significant EV charging, despite these networks being the most susceptible to voltage and thermal problems.
2. Many optimization-based charging methods are designed for static operating conditions and lack the real-time adaptability required to handle PV intermittency and dynamic EV arrivals and departures.
3. Aggregated EV demand is commonly represented as a continuous load, with limited consideration of individual vehicle scheduling or fairness among competing EVs.
4. Most research addresses voltage stability, thermal loading, or PV-induced voltage fluctuations independently rather than within a unified, network-aware optimization framework.
5. Few studies integrate a network-level optimization algorithm with an intelligent vehicle-level charging controller, highlighting the need for a coordinated hierarchical charging strategy.

The strategy proposed in this thesis responds to each of these points. It is demonstrated on a deliberately weakened high PV IEEE 33 bus feeder; it re-optimizes at every 15 minute interval on the actual network state; it resolves the decision to individual vehicles through a fuzzy priority that includes fairness; and it co-ordinates voltage, fluctuation, and thermal objectives within a two-layer architecture that joins system level optimization to vehicle level inference.

Chapter 3

3. METHODOLOGY

3.1. System Modeling and Problem Formulation

This section presents the system model used throughout the thesis. It describes the modified IEEE 33 bus distribution network, domestic load and PV generation profiles, EV charging model, and the aggregator that groups vehicles into controllable charging stations. The load flow method used to determine the network operating state is also presented. Based on these models, the coordinated charging problem is formulated as a constrained optimization problem by defining the objective function, decision variables, and operational constraints. The mathematical models developed in this chapter form the basis of the MATLAB implementation and subsequent simulation results.

3.1.1. Distribution Network Model

The IEEE 33-bus radial distribution system, originally developed by Baran and Wu [18], is adopted as the test network for this study. The system consists of a slack bus at bus 1 and thirty-two additional buses connected through thirty-two radial branches. Owing to its radial structure, each bus receives power through a single path, resulting in increasing voltage drops toward electrically distant buses. The feeder operates at a nominal voltage of 11 kV with a 100 MVA base, allowing network parameters to be expressed in the per-unit system. Its weak radial characteristics make it well suited for evaluating coordinated EV charging strategies. The base impedance is given by (1).

$$Z_{\text{base}} = V_{\text{base}}^2 / S_{\text{base}} = 11^2 / 100 = 1.21 \, \Omega \quad (1)$$

The resistance and reactance of every branch, taken from the benchmark dataset and scaled to represent the 360 mm² AAAC feeder conductor adopted here, are expressed in per unit on this base through the standard relation where z denotes a branch resistance or reactance. The per-unit conversion is given by (2).

$$Z_{\text{pu}} = Z_{\Omega} / Z_{\text{base}} \quad (2)$$

Parameter	Value
Network topology	Modified IEEE 33-bus, radial
Nominal voltage	11 kV
Power base / base impedance	100 MVA / 1.21 Ω
Distribution transformer rating	10 MVA
Feeder conductor / ampacity	360 mm ² AAAC / 525 A
Total domestic load	9.18 MVA at 0.85 p.f.
Domestic load scaling factor, κ	2.1
PV sites (buses)	10, 17, 22, 25, 32
PV rating (per site / total)	1000 kW / 5 MW _p ($\approx$ 50% penetration)
EV aggregations (buses)	6, 14, 20, 23, 26
Chargers per aggregation	up to 40 (Total 200 $\approx$ 22% EV penetration)
Charger power rating	11 kW
Charging efficiency, η_{ch}	0.90
Voltage limits	0.94 – 1.06 p.u.
Time resolution	15 min (96 intervals per day)

Table 2: Network and component parameters of the modelled distribution system

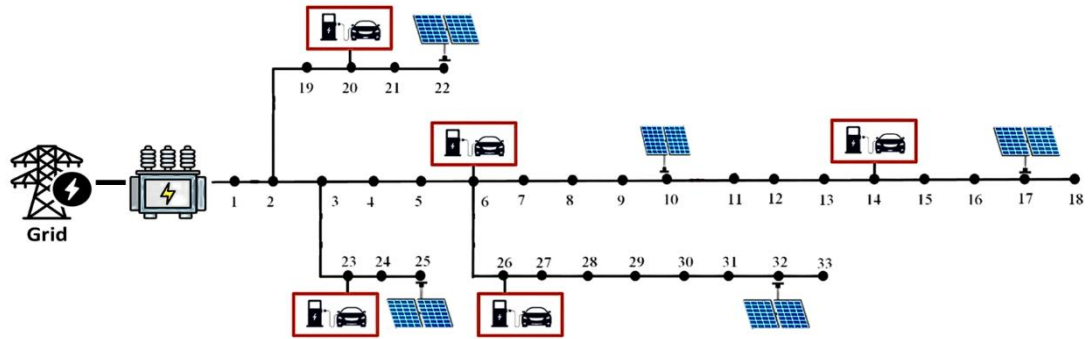


Figure 6 - Single-line diagram of the modified IEEE 33 bus test feeder

3.1.2. Weak Grid Modeling

To represent a weak distribution network, the original IEEE 33-bus loads are scaled by a factor of 2.1, increasing the total domestic demand to approximately 9.18 MVA at a power factor of 0.85. This causes the 10 MVA transformer to operate close to its rating during peak demand, making remote buses more susceptible to voltage drops

caused by additional EV charging. The chosen loading level creates a realistic operating condition where coordinated charging can demonstrate measurable improvements. It also reflects the characteristics of weak medium voltage feeders found in parts of the Sri Lankan distribution network [2].

3.1.3. Domestic Load Modeling

The domestic load is represented deterministically. Each bus carries its benchmark active and reactive demand, scaled by the weak-grid factor and then modulated through the day by a normalized load profile that captures the characteristic morning and evening peaks of residential consumption. The profile, denoted $h(t)$, is defined over the 96 fifteen-minute intervals of the day and is read directly from the load data used in the simulation. The complex demand at bus i in interval t is therefore given by (3)

$$S_{\text{load},i}(t) = \kappa (P_i + jQ_i) h(t), \quad \kappa = 2.1 \quad (3)$$

where P_i and Q_i are the benchmark active and reactive loads at the bus. The shape of the resulting daily demand, with its pronounced evening peak that later coincides with uncoordinated charging, is shown in Figure below.

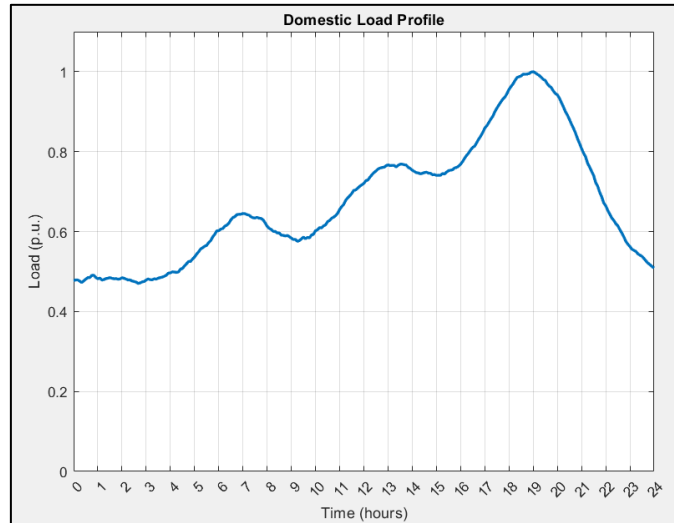


Figure 7 - Daily domestic load profile over 24 hours

3.1.4. PV Generation Modeling

PV generation is connected at five buses, each hosting a 1000 kW installation for a total of 5 MW_p, which corresponds to roughly half of the peak demand and represents a high level of penetration. The output of each site follows a normalized generation profile $p(t)$ that rises after sunrise, peaks around midday, and falls in the late afternoon. Unlike the load, the PV profile is not purely deterministic: a random noise component is superimposed on the smooth daily curve to emulate the rapid fluctuations caused by passing cloud, so that the profile reproduces the short-term variability that the controller must contend with. The generation injected at PV bus k is given by (4)

$$S_{pv,k}(t) = P_{pv} p(t), \quad P_{pv} = 1000 \text{ kW} \quad (4)$$

Because the generation is treated as a negative real load at its bus, it lifts the local voltage and reverses the branch power flow during the midday peak. The cloud affected form that is used in all simulations.

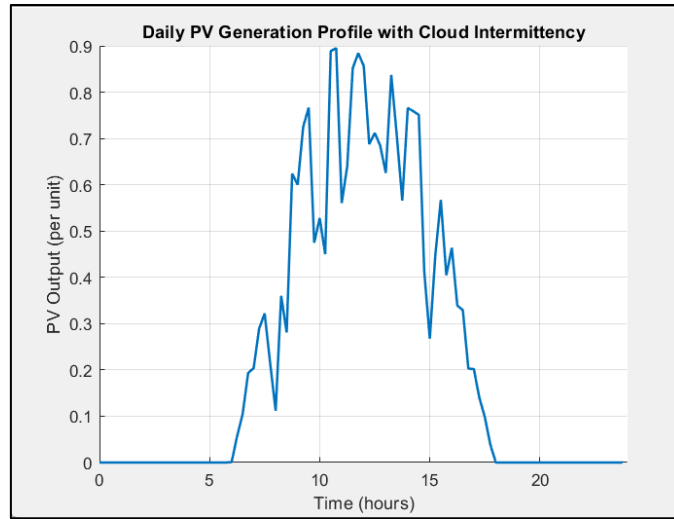


Figure 8 - Daily PV generation profile with cloud induced intermittency

3.1.5. EV Charging Model

EV Characteristics

Each EV in the simulation is characterized by its arrival and departure times, initial and target state of charge (SOC), battery capacity, and required charging energy. These parameters are generated stochastically to capture the diversity of real-world charging

behavior based on measured transaction data [8]. Consequently, the charging demand varies across aggregations and throughout the day. The required charging energy is calculated using the energy requirement equation presented later, assuming a charging efficiency of 90%.

Arrival (h)	Departure (h)	Arrival SOC (%)	Expected SOC (%)	Battery (kWh)	Required energy (kWh)
06:15	14:30	27	90	40	28
06:45	13:45	43	95	90	52
07:15	13:30	46	95	90	49

Table 3: Representative electric-vehicle charging sessions

Arrival and Departure Modeling

An EV is eligible for charging only during the period between its arrival and departure, provided its charging requirement has not yet been met. Vehicle arrival and departure times are generated to produce realistic occupancy patterns that vary throughout the day.

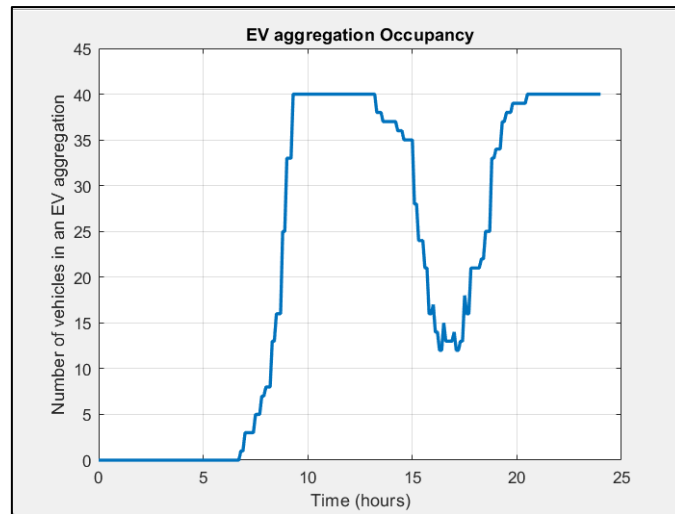


Figure 9 - Occupancy of an EV aggregation over the day

A binary presence indicator is used to determine vehicle availability during each time interval. As the number of connected vehicles changes continuously, the coordinated charging strategy must dynamically update charging decisions to accommodate the evolving demand while satisfying network and operational constraints.

Battery Model

The energy stored in a vehicle's battery is tracked through its state of charge. When a vehicle is being charged during an interval of length $\Delta t = 0.25$ h, the charger draws power $P_{ch} = 11$ kW from the grid, of which a fraction η_{ch} reaches the battery. The state of charge therefore advances according to (5)

$$SOC_i(t+1) = SOC_i(t) + (\eta_{ch} P_{ch} \Delta t / C_{bat,i}) \times 100 \quad (5)$$

where $C_{bat,i}$ is the battery capacity in kWh and the state of charge is expressed as a percentage. The state of charge is held constant in intervals when the vehicle is present but not selected for charging, and charging stops once the vehicle reaches its target.

Charging Requirements

The energy a vehicle must receive from the grid follows from the gap between its arrival and expected state of charge. Raising the state of charge from $SOC_{arr,i}$ to $SOC_{exp,i}$ adds a battery energy of $C_{bat,i}(SOC_{exp,i} - SOC_{arr,i})/100$, and because only a fraction η_{ch} of the drawn energy is stored, the grid-side energy requirement is given by (6)

$$E_{req,i} = (C_{bat,i} / \eta_{ch})(SOC_{exp,i} - SOC_{arr,i}) / 100 \quad (6)$$

This requirement defines the charging task for each vehicle. As charging proceeds, the remaining grid-side energy $E_{rem,i}$ decreases by the energy drawn in every interval the vehicle is charged, given by (7)

$$E_{rem,i}(t+1) = E_{rem,i}(t) - P_{ch} \Delta t \quad (7)$$

and the vehicle is considered satisfied once $E_{rem,i}$ reaches zero. The remaining energy is the quantity the upper-level optimizer seeks to drive down, and the priority of each vehicle in the lower layer depends on it directly.

3.1.6. EV Aggregator Model

Individual vehicles are not controlled directly. Instead, the vehicles connected at a given bus are grouped into an aggregation that the controller treats as a single resource, an arrangement that keeps the control problem tractable as the fleet grows. Each of the five aggregations exposes two quantities to the upper control layer in every interval. The

first is the number of vehicles that are present and still need energy, which sets the maximum number of chargers the aggregation could usefully activate, given by (8)

$$N_{\text{avail},s}(t) = | \{ i \in \Omega_s : t_{\text{arr},i} \leq t < t_{\text{dep},i}, E_{\text{rem},i}(t) > 0 \} | \quad (8)$$

where Ω_s is the set of vehicles assigned to station s . The second is the aggregate energy still owed to the vehicles present at the station, and its sum over all stations gives the total backlog that the optimizer works to reduce, given by (9)

$$E_{\text{rem},s}(t) = \sum_i E_{\text{rem},i}(t), \quad E_{\text{rem}}^{\text{tot}}(t) = \sum_s E_{\text{rem},s}(t) \quad (9)$$

These two signals form the interface between the network-level decision and the vehicle-level decision. The aggregator receives a permitted number of active chargers from the upper layer and is responsible, through the fuzzy controller for deciding which of its present vehicles actually charge.

3.1.7. Load Flow Analysis

The state of the network in any interval, namely the bus voltages, the branch currents, the transformer loading, and the losses, is obtained by load flow. Because the feeder is radial and has a high resistance-to-reactance ratio, the Newton–Raphson and Gauss–Seidel methods commonly used for transmission systems are poorly suited to it, and the backward/forward sweep method is used instead [19]. The method exploits the radial structure to solve the network iteratively without forming or inverting an admittance matrix.

At the start of each iteration the load current at every bus is computed from the net injected power and the current voltage estimate, given by (10)

$$I_i = (S_{\text{net},i} / V_i)^* \quad (10)$$

where the net injection combines the scaled domestic load, the electric-vehicle charging load, and the photovoltaic generation at the bus, given by (11)

$$S_{\text{net},i}(t) = S_{\text{load},i}(t) + N_i(t) P_{\text{ch}} - S_{\text{pv},i}(t) \quad (11)$$

with $N_i(t)$ the number of active chargers at the bus, non-zero only at the five aggregation buses. In the backward sweep, branch currents are accumulated from the extremities

toward the source, each branch carrying the load current of its downstream bus plus the currents of all branches fed from it, given by (12)

$$J_b = I_{L,b} + \sum_{m \in D(b)} J_m \quad (12)$$

where $D(b)$ is the set of branches immediately downstream of branch b . In the forward sweep, voltages are updated from the slack bus outward using the branch voltage drop, given by (13)

$$V_r = V_s - Z_b J_b \quad (13)$$

with V_s and V_r the sending- and receiving-end voltages of the branch and Z_b its impedance. The two sweeps repeat until the largest change in bus voltage between successive iterations falls below a tolerance, given by (14)

$$\max_i |V_i^{(k)} - V_i^{(k-1)}| < \epsilon, \quad \epsilon = 10^{-6} \quad (14)$$

which is reached in a handful of iterations for this network. The converged solution yields the bus voltages in per unit, the apparent power drawn at the slack bus, which gives the transformer loading $|S_1|$, the main-feeder current in amperes, and the branch losses. From the same solution a voltage-stability index is evaluated at every bus to gauge how close each is to voltage collapse, following the formulation for radial networks [6], [17], given by (15)

$$VSI_r = V_s^4 - 4(P_r X - Q_r R)^2 - 4(P_r R + Q_r X)V_s^2 \quad (15)$$

where P_r and Q_r are the active and reactive power flowing to the receiving bus and R and X are the branch resistance and reactance. A value near unity indicates a healthy bus and a value approaching zero a bus close to instability. The load flow is the computational engine of the whole study: it is called once per interval to record the operating point, and many times within each interval by the optimizer to evaluate candidate charging decisions.

3.1.8. Problem Formulation

With the model established, the coordination task can be stated as a constrained optimization solved once in every interval. The aim is to choose how many chargers each aggregation operates so that the network stays within its voltage and thermal limits

and voltage fluctuation is suppressed, while serving as much of the outstanding charging demand as possible.

Decision Variables

The decision in interval t is the vector of active charger counts across the five aggregations, given by (16)

$$N(t) = [N_1, N_2, N_3, N_4, N_5] \quad (16)$$

Each N_s is a non-negative integer that cannot exceed the number of vehicles available at that station, so the search space is bounded directly by the aggregator signal of Equation (3.8). Choosing the charger count, rather than a continuous power level, keeps the decision discrete and physically meaningful, since a charger is either switched on at its rated power or left off.

Objective Function

The objective combines four weighted terms, each expressing one requirement of the coordination problem, given by (17)

$$J = w_1 J_1 + w_2 J_2 + w_3 J_3 + w_4 J_4 \quad (17)$$

The first term penalizes any departure of a bus voltage from the permitted band, summed over all buses, and is given the dominant weight so that voltage limits behave almost as a hard constraint, given by (18)

$$J_1 = \sum_i [\max(V_i - V_{\max}, 0) + \max(V_{\min} - V_i, 0)] \quad (18)$$

with $V_{\min} = 0.94$ and $V_{\max} = 1.06$ p.u. The second term suppresses voltage fluctuation at bus 18, the bus at the end of the main feeder that exhibits the largest voltage excursions, by penalizing its deviation from the average of the three preceding intervals. This term is active only during the midday window of intervals 30 to 60, when photovoltaic intermittency is most pronounced, given by (19)

$$J_2 = 1000 (V_{18}(t) - V_{18,\text{avg}}(t))^2 \quad \text{for } 30 \leq t \leq 60, \text{ else } 0 \quad (19)$$

where $V_{18,\text{avg}}(t) = [V_{18}(t-1) + V_{18}(t-2) + V_{18}(t-3)] / 3$ is the three-interval rolling mean. The third term drives the charging backlog down by rewarding the energy served in the interval, expressed as the total remaining energy net of what the chosen chargers would deliver, given by (20)

$$J_3 = E_{\text{rem}}^{\text{tot}}(t) - \sum_s N_s(t) P_{\text{ch}} \Delta t \quad (20)$$

and the fourth term enforces thermal security by penalizing, quadratically, any transformer loading above its 10 MVA rating, given by (21)

$$J_4 = (\max(|S_1(t)| - S_T, 0))^2, \quad S_T = 10 \text{ MVA} \quad (21)$$

The weights set the priority among these aims. Voltage compliance and thermal security dominate, fluctuation suppression is significant during the PV window, and the backlog term, though small in weight, is what biases the optimizer toward serving demand whenever doing so does not threaten the network. The values used are listed in Table 3.3.

Term	Requirement	Weight
J_1	Voltage-limit violation (all buses)	10^9
J_2	Voltage fluctuation at bus 18 (PV window)	10^6
J_3	Remaining charging-energy backlog	10^{-3}
J_4	Transformer overload above 10 MVA	10^6

Table 4: Weighting of the objective-function terms

Constraints

The optimization is subject to the operating limits of the network and the availability of the vehicles. The bus voltages must remain within the permitted band, the transformer loading must not exceed its rating, the main-feeder current must stay within the conductor ampacity, and the charger count at each station is bounded below by zero and above by the number of available vehicles, given by (22) and (23)

$$V_{\min} \leq V_i \leq V_{\max} \quad \forall i; \quad |S_1| \leq S_T; \quad |I_1| \leq I_{\text{amp}} \quad (22)$$

$$0 \leq N_s \leq N_{\text{avail},s}(t), \quad N_s \in \mathbb{Z}^+, \quad s = 1, \dots, 5 \quad (23)$$

with $I_{\text{amp}} = 525$ A. The voltage and thermal limits are handled as strongly weighted penalties within the objective rather than as strict mathematical constraints, which lets the optimizer settle on the least bad option in the rare intervals when no decision can keep every quantity perfectly within bounds, while the charger count limits are enforced strictly through the bounds of the search space. The feeder current limit is monitored as an operating constraint and reported in the results.

3.2. Proposed Coordinated Charging Strategy

This section presents the proposed coordinated charging strategy for solving the optimization problem formulated in the previous chapter. The framework consists of two hierarchical control layers. At the upper level, PSO determines the charging capacity that the network can safely support, while the lower-level fuzzy logic controller allocates available chargers to individual EVs based on their charging priority. The chapter describes the overall architecture, the operation of each control layer, and the complete coordination algorithm, following the actual MATLAB implementation used in this research.

3.2.1. Overall Control Architecture

Coordinating a large number of vehicles on a constrained network involves two questions of very different character. The first is a network question: given the present load, generation, and voltage profile, how many chargers can each aggregation safely operate without driving the feeder outside its limits? The second is an allocation question: among the vehicles connected at an aggregation, which ones should receive those chargers? Answering both at once, for every vehicle and every interval, would be a large mixed-integer problem coupled to a nonlinear load flow, and solving it repeatedly in real time would be impractical.

The proposed strategy avoids this by splitting the problem into two layers that match the two questions [4]. The upper layer is a particle swarm optimizer that works at the level of aggregations. It decides the number of active chargers at each of the five stations, evaluating candidate decisions against the network state through the load flow, and returns the charger-count vector that best balances voltage stability, thermal security, and demand service. The lower layer is a fuzzy controller that works at the level of individual vehicles. Taking the charger count allotted to a station as a hard cap, it ranks the vehicles present at that station by a priority score and switches on only the highest-priority ones. Because the upper layer never decides which specific vehicle charges, and the lower layer never exceeds the capacity the upper layer has authorized, the grid constraint is respected by construction while the per-vehicle decision remains fast and adaptive. The interaction between the two layers is shown in Figure 4.1.

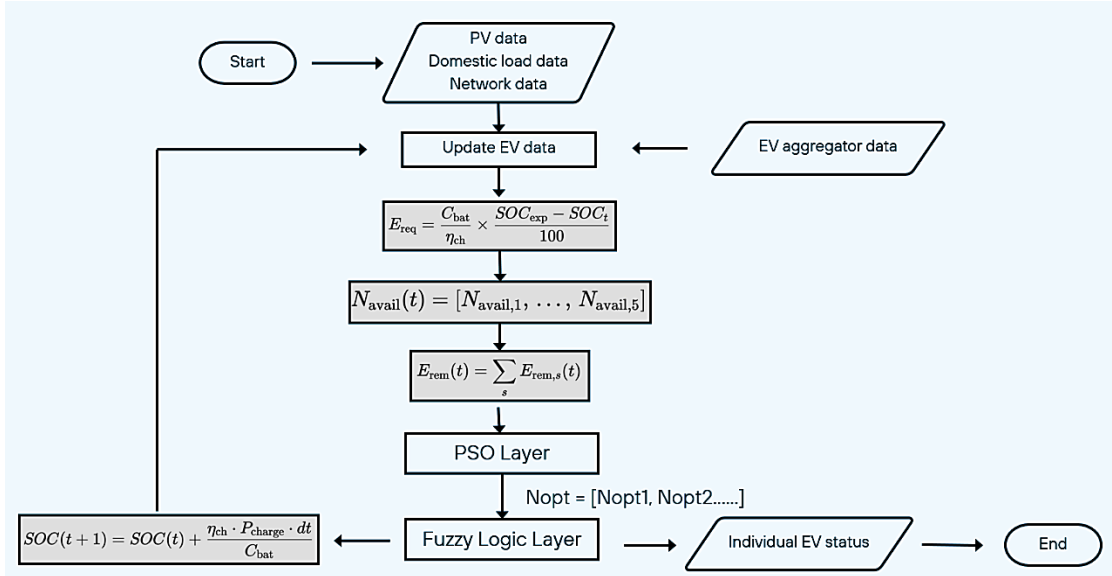


Figure 10 - Overall two layer coordinated charging architecture

3.2.2. PSO Based Aggregator Scheduling

The upper layer determines, for the current interval, how many chargers each aggregation should operate. Particle swarm optimization is well suited to this task because the objective is non-convex and is evaluated through a nonlinear load flow that cannot be differentiated in closed form, and because the method searches the decision space directly without requiring gradients [10].

Particle Representation

Each particle encodes a complete candidate decision as a five-element vector, one entry per aggregation, given by (24)

$$x = [N_1, N_2, N_3, N_4, N_5] \quad (24)$$

where N_s is the number of active chargers proposed for station s . A particle therefore occupies a point in a five-dimensional space bounded below by zero and above by the number of vehicles currently available at each station. The swarm is initialized by scattering twenty particles uniformly within these bounds. Because a fractional number of chargers has no physical meaning, each particle position is rounded to the nearest integer before it is evaluated, although the particles themselves move through continuous space so that the search remains smooth.

Fitness Function

$$J = w_1 J_1 + w_2 J_2 + w_3 J_3 + w_4 J_4 \quad (25)$$

The objective function is given by (25). Each candidate solution is evaluated using a complete load flow analysis, allowing network voltages and transformer loading to be assessed under actual operating conditions. Although this network-aware evaluation increases computational effort, it enables the optimizer to make accurate charging decisions based on the true system state.

Constraints

The decision is bounded by the availability of vehicles, expressed as (26)

$$0 \leq N_s \leq N_{\text{avail},s}(t), \quad N_s \in \mathbb{Z}^+ \quad (26)$$

Particle positions are constrained within predefined bounds using a reflection mechanism that prevents them from leaving the feasible search space. Network voltage and thermal limits are incorporated into the PSO objective function as high-penalty terms rather than hard constraints. This approach simplifies the optimization process while strongly discouraging solutions that violate operational network limits.

PSO Algorithm

At each iteration every particle updates its velocity from three influences: its present motion, its attraction toward the best position it has personally found, and its attraction toward the best position found by the whole swarm, given by (27)

$$v_s^{(k+1)} = w v_s^{(k)} + c_1 r_1 (p_s - x_s^{(k)}) + c_2 r_2 (g_s - x_s^{(k)}) \quad (27)$$

where p_s is the particle's personal-best position, g_s the global-best position, r_1 and r_2 are random numbers drawn afresh each step, and $c_1 = 1.5$ and $c_2 = 2.0$ weight the cognitive and social pulls. The velocity is limited to ± 0.2 of each dimension's range to prevent the swarm from overshooting, and the position then advances by (28)

$$x_s^{(k+1)} = x_s^{(k)} + v_s^{(k+1)} \quad (28)$$

The inertia weight begins at unity and is reduced geometrically after each iteration, given by (29)

$$w^{(k+1)} = 0.95 w^{(k)} \quad (29)$$

so that the swarm explores broadly at first and settles into a finer search as it converges [20]. The personal and global bests are refreshed whenever a particle improves on them, and after forty iterations the global best, rounded to integers, is returned as the charger-count vector N_{opt} for the interval.

Parameter	Value
Swarm size (particles)	20
Maximum iterations	40
Particle dimension	5 (one per aggregation)
Inertia weight, w (initial)	1.0
Inertia damping ratio	0.95 per iteration
Cognitive coefficient, c_1	1.5
Social coefficient, c_2	2.0
Velocity limit	$\pm 0.2 \times (\text{per-station range})$
Boundary handling	Reflect velocity and clamp position

Table 5: Particle swarm optimization parameters

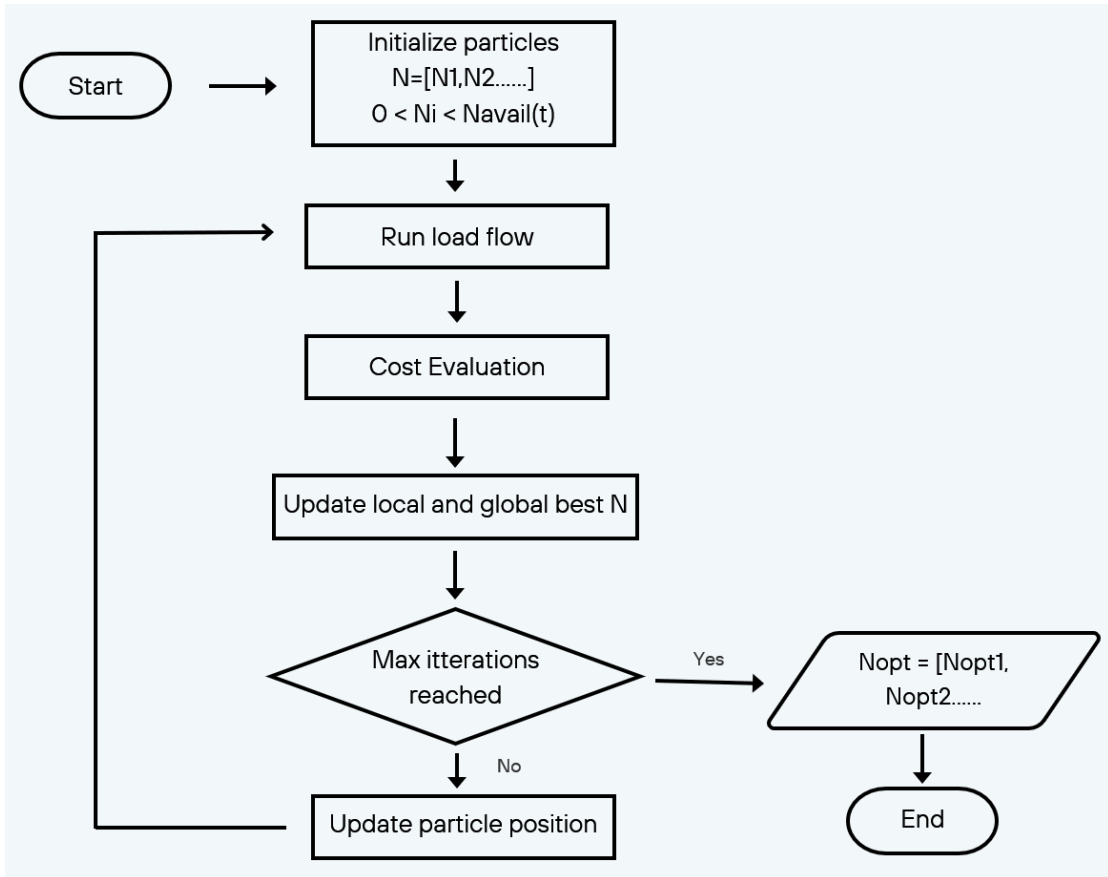


Figure 11 - Flowchart of the PSO based aggregator scheduling layer

3.2.3. Fuzzy Logic-Based Charger Allocation

Once the upper layer fixes how many chargers a station may use, the lower layer decides which vehicles receive them. This decision is made by a Mamdani fuzzy inference system, chosen because it expresses the allocation policy as transparent linguistic rules, handles the imprecision in the inputs gracefully, and is cheap enough to evaluate for every connected vehicle at every interval.

Controller Architecture

The fuzzy logic controller evaluates each vehicle independently using three input variables and produces a single charging priority score. This vehicle-by-vehicle approach keeps the controller simple, scalable, and computationally efficient. Since the fuzzy inference system is created only once at the beginning of the simulation, its initialization cost is negligible during real-time operation.

Input Variables

The three inputs are computed for each vehicle i in the current interval t . The first is the energy demand, the fraction of the required energy that is still outstanding, so that a vehicle with most of its charging still to do scores near one. It is given by (30).

$$ed_i = E_{\text{rem},i} / E_{\text{req},i} \in [0, 1] \quad (30)$$

The second is the remaining time, the number of intervals before the vehicle departs, which measures urgency, since a small value means the vehicle must be served soon or not at all. It is given by (31).

$$rt_i = t_{\text{dep},i} - t \quad (31)$$

The third is the fairness index, the number of intervals the vehicle has already waited since arriving, which rises the longer a vehicle goes without being charged and prevents low-demand or low-urgency vehicles from being indefinitely overlooked. It is given by (32).

$$ft_i = t - t_{\text{arr},i} \quad (32)$$

Energy demand is defined on the unit interval, while remaining time and fairness are defined over the ninety-six intervals of the day.

Membership Functions

Each input and the output are partitioned into three fuzzy sets, labelled Low, Medium, and High. The two outer sets use sigmoidal membership functions and the central set a generalized bell function, a shouldered arrangement that saturates smoothly at the extremes while giving the middle range a rounded, gradual response [21]. The sigmoidal and bell functions are given by (33) and (34)

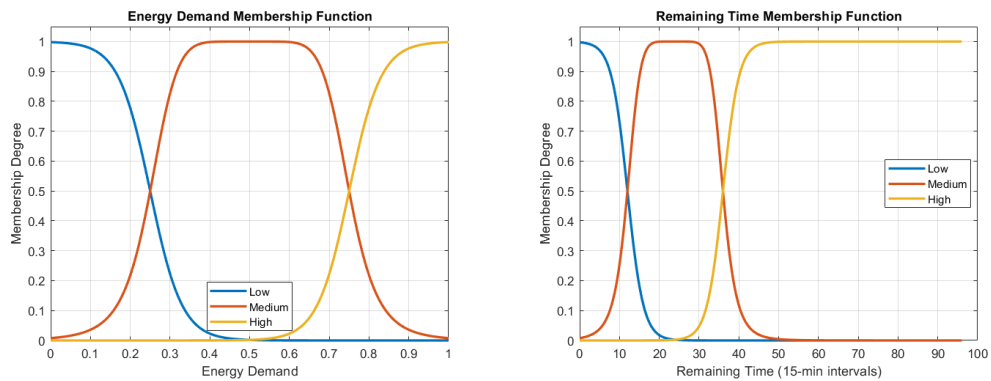
$$\text{sigmf}(x; a, c) = 1 / (1 + e^{-a(x-c)}) \quad (33)$$

$$\text{gbellmf}(x; a, b, c) = 1 / (1 + |(x-c)/a|^{2b}) \quad (34)$$

where c locates the function, a sets its slope or width, and b the sharpness of the bell. The parameters used for every set, taken directly from the implemented inference system

Variable (range)	Term	Type	Parameters
Energy Demand [0, 1]	Low	sigmf	$a = -25, c = 0.25$
	Medium	gbellmf	$a = 0.25, b = 3.5, c = 0.5$
	High	sigmf	$a = 25, c = 0.75$
Remaining Time [0, 96]	Low	sigmf	$a = -0.5, c = 12$
	Medium	gbellmf	$a = 12, b = 3.5, c = 24$
	High	sigmf	$a = 0.5, c = 36$
Fairness Index [0, 96]	Low	sigmf	$a = -0.5, c = 12$
	Medium	gbellmf	$a = 12, b = 3.5, c = 24$
	High	sigmf	$a = 0.5, c = 36$
Priority [0, 1]	Low	sigmf	$a = -25, c = 0.3$
	Medium	gbellmf	$a = 0.2, b = 3.5, c = 0.5$
	High	sigmf	$a = 25, c = 0.7$

Table 6: Membership-function definitions for the fuzzy inputs and output



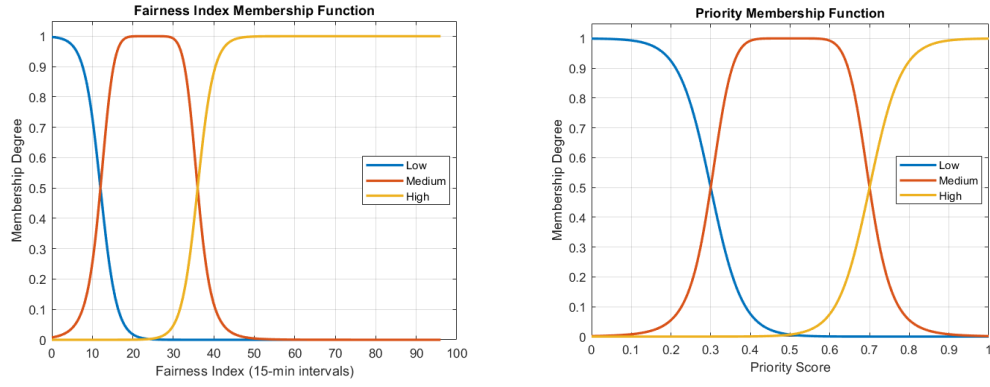


Figure 12 - Membership functions of the fuzzy inputs and output

Rule Base

The policy is encoded as a complete set of twenty-seven rules, one for every combination of the three inputs at their three linguistic levels. The rules express a consistent priority logic: a vehicle with high outstanding demand and little time left is urgent and receives high priority, a nearly full vehicle with plenty of time receives low priority, and the fairness index lifts the priority of vehicles that have waited a long time so that they are not starved.

Fairness Index = Low (recently arrived)			
Energy Demand ↓ / Time →	Low	Medium	High
High	High	Medium	Low
Medium	Medium	Low	Low
Low	Low	Low	Low
Fairness Index = Medium			
Energy Demand ↓ / Time →	Low	Medium	High
High	High	High	Medium
Medium	High	Medium	Low
Low	Medium	Low	Low
Fairness Index = High (long wait)			
Energy Demand ↓ / Time →	Low	Medium	High
High	High	High	High
Medium	High	Medium	Medium
Low	Medium	Low	Low

Table 7: Fuzzy rule base

Defuzzification

The rules are combined by Mamdani inference. For a given vehicle, the membership of each input in its three sets is found, the antecedents of each rule are combined by the fuzzy AND, taken as the minimum, and the rule's strength clips its consequent set. The clipped consequents of all rules are aggregated, and the resulting fuzzy set is converted back to a single number by the centroid method, given by (35)

$$P_i = \int z \mu(z) dz / \int \mu(z) dz \quad (35)$$

where $\mu(z)$ is the aggregated output membership and $P_i \in [0, 1]$ is the crisp priority of vehicle i . The centroid is used because it reflects the whole shape of the aggregated set and produces a smooth, well-graded priority that discriminates cleanly between vehicles of similar standing.

Charging Priority Calculation

In each interval, the three inputs are formed for every available vehicle at a station and evaluated through the inference system in a single batch, giving a priority score for each. The vehicles are then sorted in descending order of priority, and the highest-scoring ones are switched on up to the cap set by the upper layer, given by (36)

$$N_s = \min(N_{\text{opt},s}, N_{\text{avail},s}) \quad (36)$$

The fuzzy controller allocates the available chargers without exceeding the limit determined by the PSO layer. Selected EVs are charged, while the remaining vehicles wait and gain higher fairness scores, increasing their future priority. This approach balances energy demand, urgency, and fairness while ensuring compliance with network constraints [16].

3.2.4. Complete Coordination Algorithm

The two layers are combined in a single procedure that advances the simulation interval by interval. At the start of each interval the EV states are updated and the aggregator signals are formed; the optimizer is then run to fix the charger counts; a final load flow records the resulting network state; and the fuzzy controller allocates the authorized chargers to individual vehicles.

3.2.5. Computational Workflow

At each 15-minute interval, vehicle arrivals and departures are updated before the PSO determines the optimal number of active chargers. This stage is computationally intensive, requiring numerous load flow evaluations. However, the efficient backward/forward sweep load flow method keeps the overall computation practical for real-time coordinated charging.

After the optimal charger counts are determined, a final load flow is performed to record the network state and update the voltage profile for the next interval. The fuzzy controller then assigns chargers to individual EVs with minimal computational effort. Vehicle energy levels and state of charge are updated before recording the required performance metrics. Repeating this process over all 96 time intervals generates the daily charging profile, enabling a comparison between coordinated and uncoordinated charging strategies in the following chapter.

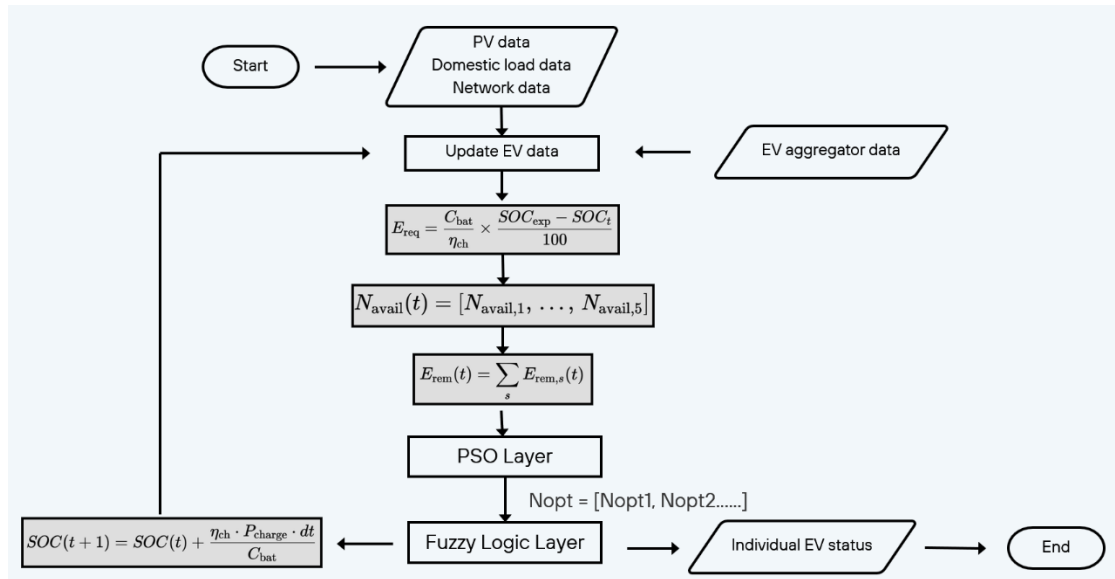


Figure 13 - Computational workflow of the coordinated charging strategy across one interval

Chapter 4

4. SIMULATION SETUP

4.1. Introduction

This chapter describes the simulation environment, input data, and evaluation methodology used in this research. It presents the network parameters, load, PV, and EV datasets, defines the coordinated and uncoordinated charging scenarios, and introduces the performance metrics used for comparison. These details ensure that the simulation can be reproduced and the reported results independently verified.

4.2. Simulation Environment

The simulation is implemented in MATLAB R2023a using custom scripts for the load flow, PSO and supporting calculations, ensuring complete control over the modelling process. The only toolbox used is the Fuzzy Logic Toolbox, which implements the Mamdani fuzzy controller [22]. Two main scripts are executed sequentially. The first performs the uncoordinated charging simulation and stores the baseline results, while the second executes the coordinated charging strategy for comparison. Both simulations use the same IEEE 33-bus network, PV and load profiles, EV dataset, and 24-hour simulation horizon divided into ninety-six 15-minute intervals. This ensures that any observed performance differences are solely due to the proposed coordinated charging strategy.

4.3. Network Parameters

The simulation uses the modified IEEE 33 bus radial distribution network presented in Chapter 3. The feeder operates at 11 kV with a 100 MVA base and is supplied through a 10 MVA transformer. To represent a weak grid, the domestic load is scaled to approximately 9.18 MVA at a power factor of 0.85. Five PV stations with a combined capacity of 5 MWp and five EV charging aggregations are integrated into the network. Bus voltages are maintained within the range of 0.94–1.06 p.u. The network configuration remains unchanged throughout the simulations, allowing the analysis to focus solely on the effects of different charging strategies.

4.4. Load and PV Profiles

The simulation uses normalized daily load and PV generation profiles divided into 96 fifteen-minute intervals. The domestic load follows a typical residential pattern, with peak demand occurring during the evening. The PV profile increases after sunrise, peaks around midday, and includes short-term fluctuations to represent cloud-induced intermittency. Using this realistic PV profile enables the coordinated charging strategy to be evaluated under varying generation conditions. The analysis is based on a single representative day, while seasonal and multi-day variations are beyond the scope of this study.

4.5. EV Dataset

The simulation includes five EV aggregations, each serving up to 40 vehicles, resulting in a total fleet of 200 EVs. Vehicle characteristics, including arrival and departure times, battery capacity, state of charge, and energy demand, are generated stochastically based on observed charging behavior [8]. Separate datasets are used for each aggregation and loaded at the start of the simulation.

Station	Bus	No. of EVs	Dataset file
Aggregation 1	6	40	EV_Data_1.xlsx
Aggregation 2	14	40	EV_Data_2.xlsx
Aggregation 3	20	40	EV_Data_3.xlsx
Aggregation 4	23	40	EV_Data_4.xlsx
Aggregation 5	26	40	EV_Data_5.xlsx
Total		200	

Table 8: Electric-vehicle aggregations and their datasets

Each vehicle's required energy is derived from its arrival and target states of charge and its battery capacity through the charging-requirement relation of Chapter 3, evaluated at a charging efficiency of 0.9.

4.6. Simulation Scenarios

Two scenarios are simulated on the same network and the same vehicle population, so that they differ only in how charging is controlled.

4.6.1. Uncoordinated Charging

In the uncoordinated charging scenario, every EV starts charging immediately upon arrival and continues at full charging power until reaching its target state of charge or departure time. Charging decisions are made without considering network conditions, causing EV demand to coincide with the evening residential peak. This unmanaged charging behavior represents the worst case operating condition for the distribution network and serves as the baseline for evaluating the effectiveness of the proposed coordinated charging strategy.

4.6.2. Coordinated Charging

In the coordinated charging scenario, the proposed two-layer control framework manages EV charging at every 15-minute interval. The PSO determines the optimal number of active chargers at each aggregation based on current network conditions, while the fuzzy logic controller prioritizes individual vehicles for charging. Charging demand is shifted to periods with available network capacity, ensuring voltage and thermal limits are maintained. The same EV fleet and charging requirements are used as in the uncoordinated case, allowing a fair comparison of the two charging strategies.

4.7. Performance Evaluation Metrics

The two scenarios are compared through a set of metrics that together capture voltage stability, thermal security, and the quality of service delivered to the vehicles. The voltage performance is judged from the bus voltage profiles over the day and, in particular, from the lowest voltage reached at the most vulnerable bus, given by (37)

$$V_{\min} = \min_{i, t} V_i(t) \quad (37)$$

a value that must stay above 0.94 p.u. for the network to be compliant. Voltage stability is quantified more finely by the voltage-stability index of Chapter 3, summarized for each bus by its worst value over the day [6], given by (38)

$$VSI_{\min, i} = \min_t VSI_i(t) \quad (38)$$

with larger values indicating a healthier, more stable bus. Thermal security is assessed from the transformer loading and the feeder current. The transformer peak is the largest

apparent power drawn at the supply over the day, which must not exceed the 10 MVA rating, given by (39)

$$S_{T,peak} = \max_t | S_1(t) | \quad (39)$$

and the feeder loading during the evening peak is expressed as a percentage of the conductor ampacity, averaged over the peak window, given by (40)

$$L_{np} = (100 / N_{np} I_{amp}) \sum_{t \in \text{peak}} | I_1(t) | \quad (40)$$

The quality of service to the vehicles is captured by the energy-demand satisfaction, the share of the total requested energy that is actually delivered by the time the vehicles depart, given by (41)

$$\eta_{sat} = 100 \times \sum_i (E_{req,i} - E_{rem,i}) / \sum_i E_{req,i} \quad (41)$$

Network efficiency and cost are measured by the total energy lost in the branches over the day and by the charging cost, the latter evaluated against a time-of-use tariff, given by (42)

$$E_{loss} = \sum_t \sum_b P_{loss,b}(t) \Delta t, \quad C = \sum_t \pi(t) P_{tot}(t) \Delta t \quad (42)$$

where $P_{tot}(t)$ is the total charging power and $\pi(t)$ is the tariff in force. The tariff follows the time-of-use structure of the Sri Lankan distribution utility, with cheaper off-peak energy overnight and the highest rate during the evening peak [2] Finally, the benefit to the network is summarized by the peak-shaving achieved, the reduction in transformer peak loading between the two scenarios, given by (43)

$$\Delta S_{peak} = S_{T,peak}^{unco} - S_{T,peak}^{co} \quad (43)$$

Period	Time of day	Intervals	Tariff (LKR/kWh)
Off-peak	22:15 – 05:15	90–96, 1–21	20
Day	05:15 – 18:15	22–73	30
Peak	18:15 – 22:15	74–89	55

Table 9: Time-of-use tariff blocks used to evaluate charging cost

Chapter 5

5. RESULTS AND DISCUSSION

This chapter presents and discusses the simulation results obtained for both uncoordinated and coordinated EV charging. The uncoordinated case is analyzed first to establish the baseline performance of the weak distribution network, followed by the proposed coordinated charging strategy under identical conditions. The results are compared using the performance metrics defined in the previous chapter. Finally, the findings are interpreted, key modelling assumptions are discussed, and the limitations of the proposed approach are highlighted.

5.1. Uncoordinated Charging Performance

5.1.1. Voltage Profile

The voltage profile under uncoordinated charging clearly demonstrates the vulnerability of the weak distribution feeder. While the voltage at bus 5 remains close to its nominal value throughout the day, the voltage at the electrically distant bus 18 decreases significantly during the evening charging period. As large numbers of EVs begin charging simultaneously, the voltage falls to approximately 0.92 p.u., violating the minimum allowable limit of 0.94 p.u. These results confirm that unmanaged EV charging primarily affects remote buses and can lead to unacceptable voltage conditions.

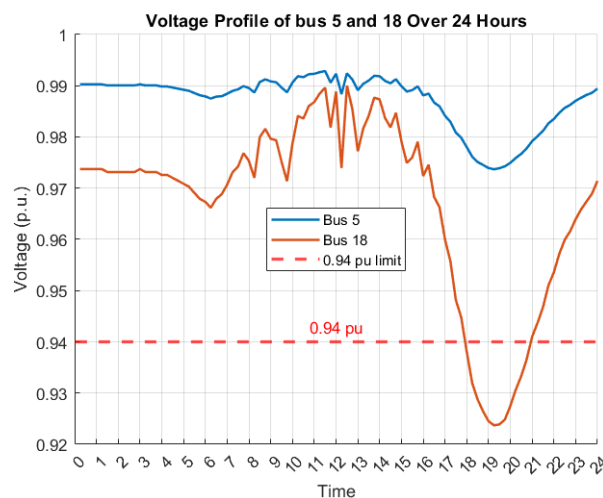


Figure 14 - Voltage profile at buses 5 and 18, uncoordinated charging

5.1.2. Transformer Loading

The transformer loading exhibits a similar trend under uncoordinated charging. During the evening peak, the loading increases to approximately 11.3 MVA, exceeding the 10 MVA transformer rating. This overload occurs simultaneously with the voltage violation, confirming that unmanaged EV charging places significant thermal and voltage stress on the distribution network.

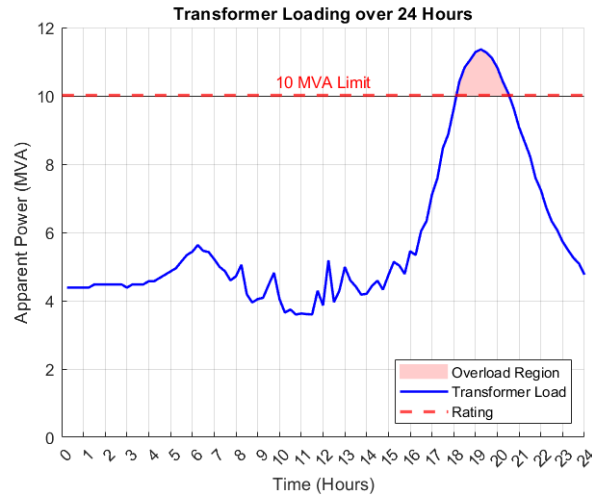


Figure 15 - Transformer loading over 24 hours, uncoordinated charging

5.1.3. Line Loading

During the evening peak, the main feeder current reaches 100.03% of its 525 A ampacity, slightly exceeding the conductor's thermal rating.

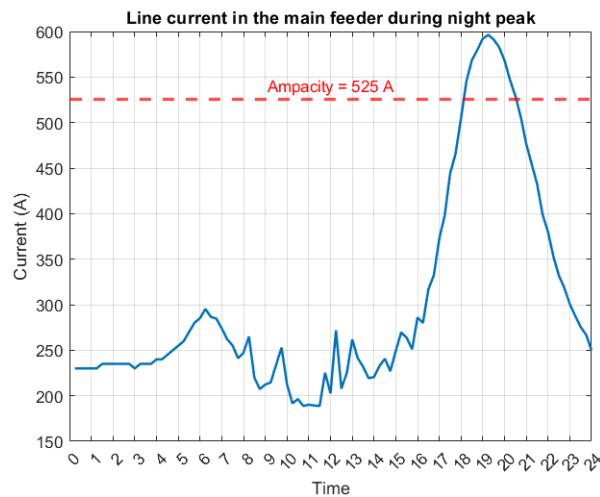


Figure 16 - Main feeder current during the night peak, uncoordinated charging

5.1.4. Power Loss

The high evening charging demand increases branch currents and network losses. Under uncoordinated charging, the total daily energy loss reaches 2,659.73 kWh, providing the baseline for evaluating the effectiveness of the coordinated charging strategy.

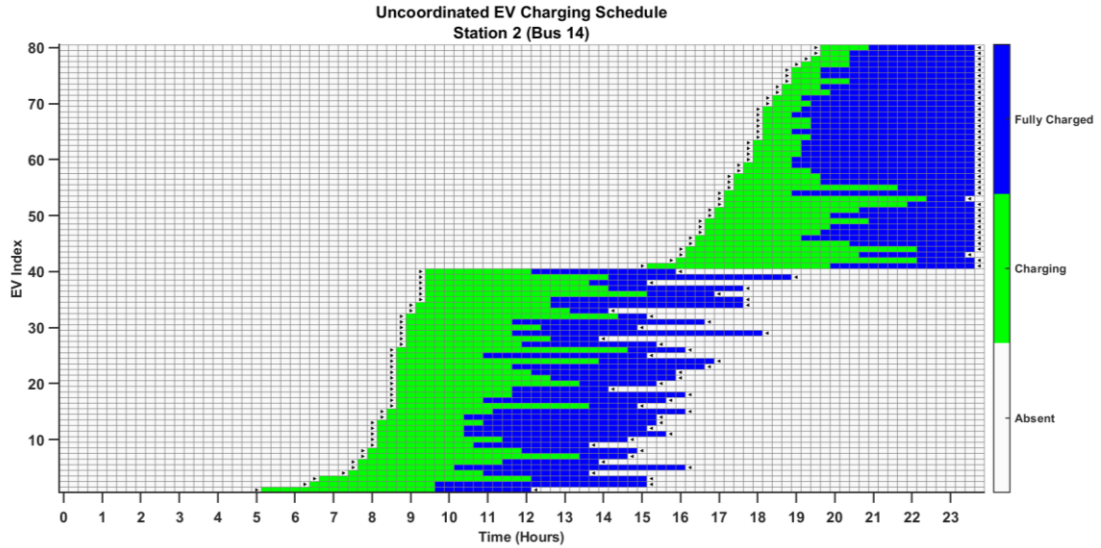


Figure 17 - EVs charging status of a selected aggregation

5.2. Coordinated Charging Performance

5.2.1. Voltage Profile

Under coordinated charging, the voltage at bus 18 remains within the allowable limit throughout the day, with a minimum value of approximately 0.94 p.u. The optimization strategy also reduces voltage fluctuations during periods of variable PV generation, improving overall voltage stability across the distribution network.

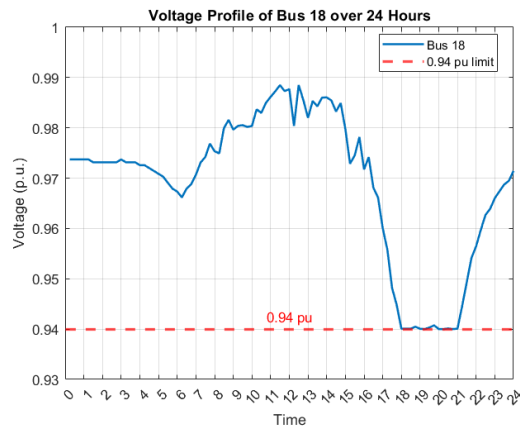


Figure 18 - Voltage profile at bus 18, coordinated charging

5.2.2. Transformer Loading

The transformer loading is held within its rating. The evening peak that reached about 11.3 MVA in the baseline is brought down to approximately 10 MVA, so the transformer is no longer overloaded. The optimizer achieves this by limiting how many chargers operate during the peak and deferring the rest, which flattens the top of the loading curve rather than letting it spike.

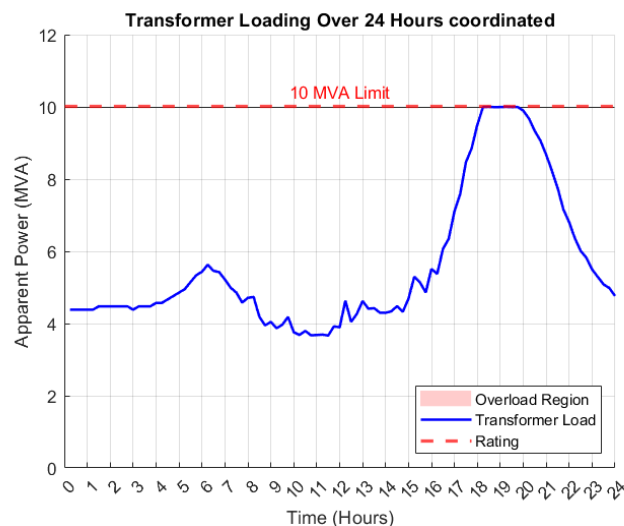


Figure 19 - Transformer loading over 24 hours, coordinated charging

5.2.3. Line Loading

Coordinated charging reduces the average evening feeder loading to 92.32% of its ampacity, keeping the feeder within thermal limits. Deferred charging is shifted to later hours when sufficient network capacity is available.

5.2.4. Charging Schedule

The coordinated charging strategy staggers EV charging based on vehicle priority rather than charging all vehicles immediately upon arrival. Priority is determined by energy demand, remaining parking time, and fairness, allowing lower-priority charging to be deferred to off-peak periods while ensuring that most vehicles achieve their required state of charge before departure.

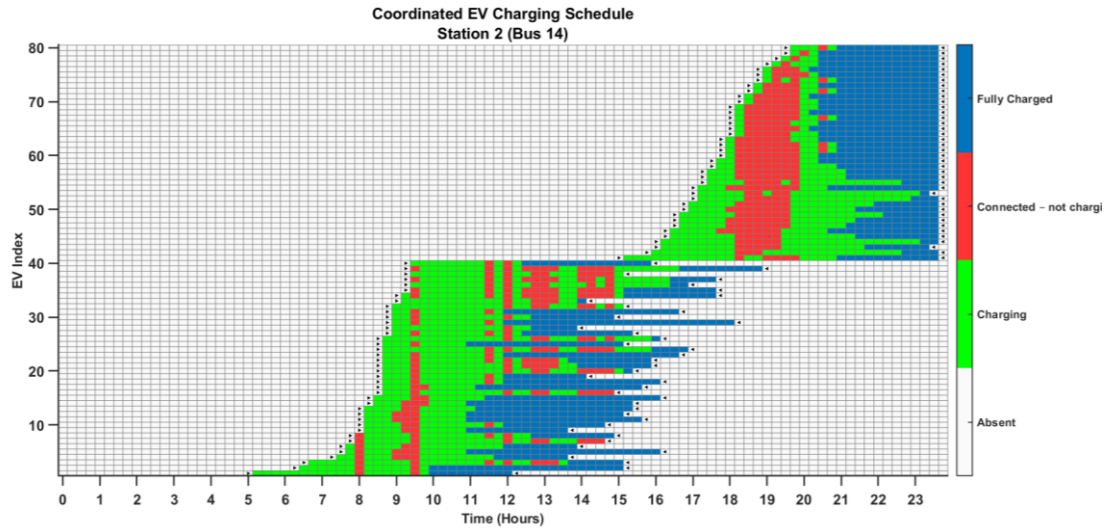


Figure 20 - EVs charging status during coordinated charging in a selected aggregation

5.3. Comparative Analysis

Placing the two scenarios side by side makes the benefit of coordination explicit. The headline metrics are collected in Table 6.1 and discussed below.

Metric	Uncoordinated	Coordinated	Change
Minimum voltage at bus 18 (p.u.)	≈ 0.92	≈ 0.94	within limit
Transformer peak loading (MVA)	≈ 11.3	≈ 10.0	≈ 1.3 shaved
Night-peak feeder loading (%)	100.03	92.32	-7.71 pts
Weakest-bus VSI (minimum)	≈ 0.73	≈ 0.78	improved
Energy-demand satisfaction (%)	≈ 100	94.08	-5.92 pts
Total energy losses (kWh)	2,659.73	2,351.54	-11.6%
Total charging cost (LKR)	414,108.75	365,854.50	-11.7%

Table 10: Comparison of uncoordinated and coordinated charging

5.3.1. Voltage Improvement

The coordinated strategy significantly improves voltage performance by increasing the minimum voltage at the weakest bus from approximately 0.92 p.u. to 0.94 p.u., eliminating voltage violations. It also reduces voltage fluctuations during periods of variable PV generation, resulting in a more stable network voltage profile.

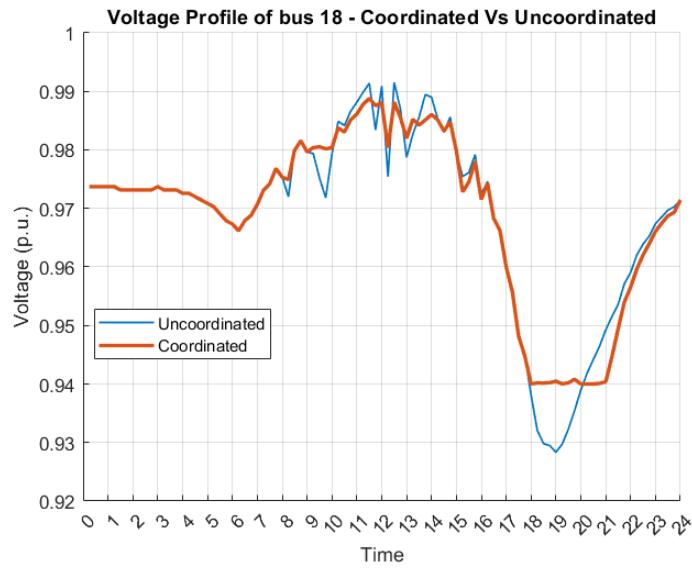


Figure 21 - Bus 18 voltage profile, coordinated versus uncoordinated

5.3.2. Thermal Improvement

Coordinated charging eliminates transformer overloading and reduces feeder loading from 100.03% to 92.32% of conductor ampacity, ensuring both thermal constraints remain within permissible operating limits.

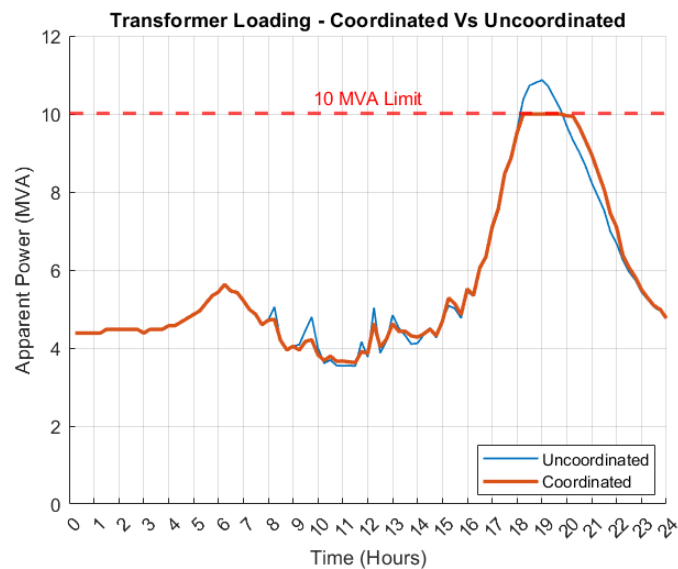


Figure 22 - Transformer loading, coordinated versus uncoordinated

5.3.3. Peak Shaving

The reduction in thermal loading is achieved through peak shaving. By limiting EV charging during the evening peak and shifting demand to off-peak hours, the strategy lowers transformer loading by approximately 1.3 MVA while delivering the required charging energy when sufficient network capacity is available.

5.3.4. Energy Satisfaction

The coordinated strategy delivers 99.94% of the total requested charging energy, leaving a small shortfall of approximately 50 kWh. This modest reduction in charging satisfaction is the trade-off for maintaining voltage and thermal limits across the distribution network.

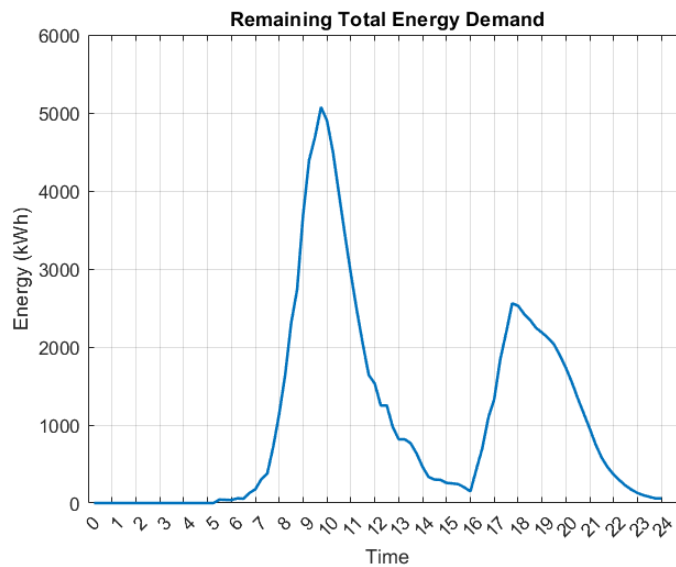


Figure 23 - Remaining total energy demand over the day, coordinated charging

5.3.5. Network Loss Reduction

By reducing peak loading and distributing charging demand more evenly, the coordinated strategy decreases daily network losses from 2,659.73 kWh to 2,351.54 kWh, an improvement of approximately 11.6%. This reduction is accompanied by higher voltage stability indices across the network, indicating an overall improvement in voltage stability [6].

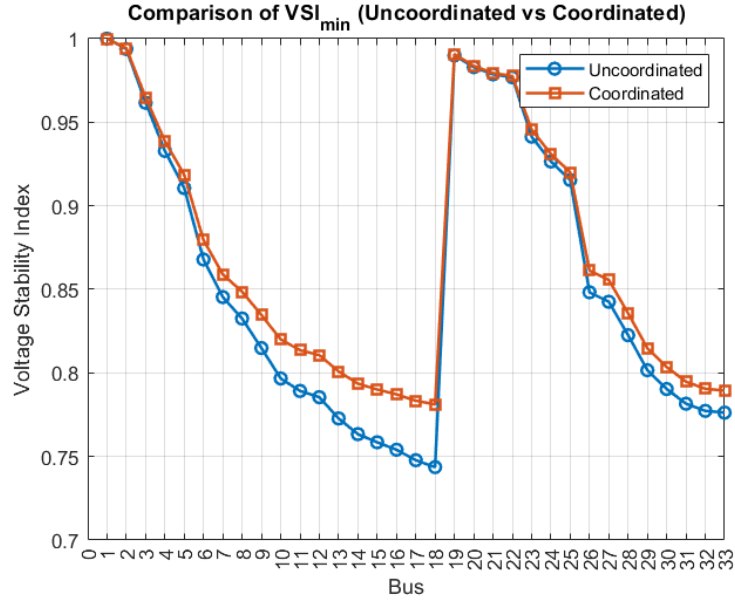


Figure 24 - Minimum voltage stability index per bus, coordinated versus uncoordinated

5.4. Discussion of Results

Overall, the simulation results demonstrate that the proposed coordinated charging strategy successfully maintains both voltage and thermal limits in a weak distribution network with high PV penetration and significant EV demand. Unlike the uncoordinated case, the proposed method prevents voltage violations, eliminates equipment overloading, reduces peak demand, and lowers network losses. These improvements are achieved through a common mechanism of shifting EV charging away from the evening peak to periods of lower network loading. Consequently, the strategy makes more effective use of existing network infrastructure, supporting the findings of previous studies that coordinated charging can postpone costly network reinforcements while improving overall system performance [4].

The coordinated charging strategy also reduces charging costs by approximately 11.7% through shifting EV charging from high-cost evening periods to lower-cost off-peak hours under the time-of-use tariff. Although cost minimization is not an explicit optimization objective, it emerges as an additional benefit of peak load management. The main trade-off is a small unmet charging demand of about 6%, which results from maintaining voltage and thermal limits during constrained operating periods. Rather than affecting the same users repeatedly, the fuzzy logic controller allocates this

shortfall fairly by prioritizing vehicles based on urgency, energy demand, and waiting time, thereby minimizing the impact on individual EV owners while preserving overall network security [3].

5.5. Sensitivity Analysis

The results presented are based on a single representative operating day with fixed simulation parameters. The performance of the proposed strategy is expected to vary with factors such as grid strength, PV penetration, EV penetration, and the weighting of the optimization objectives. A weaker grid or higher EV penetration would increase network stress, while higher PV penetration would place greater emphasis on voltage fluctuation mitigation. Similarly, changing the objective function weights would alter the balance between network protection and charging satisfaction. This study does not include a detailed sensitivity or Monte Carlo analysis, and such investigations are identified as valuable directions for future work to evaluate the robustness of the proposed strategy under a wider range of operating conditions.

5.6. Limitations of the Proposed Method

This study is based on a single modified IEEE 33-bus feeder and one representative operating day; therefore, the results demonstrate the effectiveness of the proposed strategy under the selected conditions rather than across all possible network configurations. The load and PV profiles are predefined, with variability introduced only through PV intermittency and stochastic EV behavior. In addition, the proposed controller regulates only EV charging schedules and does not consider vehicle-to-grid operation, reactive power support, network reconfiguration, or transformer tap-changing, all of which could further improve network performance. The optimization objective uses fixed weighting factors, and network constraints are enforced through penalty terms rather than strict mathematical constraints. Finally, although the coordinated strategy maintains network security, a small proportion of charging demand remains unmet during highly constrained periods. These limitations provide opportunities for future research but do not affect the main conclusions of this work.

Chapter 6

6. CONCLUSION AND FUTURE WORK

6.1. Conclusions

This thesis addressed the challenge of coordinated EV charging in a weak distribution network with high photovoltaic penetration while maintaining voltage and thermal stability. A two-layer control framework was developed, where PSO determines the number of active chargers at each aggregation based on network conditions, and a fuzzy logic controller allocates these chargers to individual vehicles according to their energy demand, remaining parking time, and fairness. This hierarchical approach enables efficient utilization of available network capacity while satisfying EV charging requirements under operational constraints.

The proposed strategy was validated using a modified IEEE 33-bus distribution feeder with five PV stations and five EV charging aggregations under weak grid conditions. Compared with uncoordinated charging, the proposed method successfully maintained voltage within acceptable limits, eliminated transformer and feeder overloading, and reduced peak demand by shifting EV charging to periods of lower network loading. It also reduced network losses by approximately 11.6% and charging costs by about 11.7%, while improving the voltage stability margin across the feeder. Although around 6% of the total charging demand remained unmet during highly constrained periods, this was necessary to preserve safe network operation. Overall, the results demonstrate that the proposed PSO–fuzzy logic framework effectively coordinates EV charging in weak, high-PV distribution networks, achieving significant improvements in network performance without requiring additional infrastructure or major modifications to the existing distribution system.

6.2. Contributions of the Research

First, a two-layer coordinated charging framework is proposed, separating network-level charger scheduling from vehicle-level charger allocation. Second, a network-aware PSO optimization is developed by embedding load flow calculations directly into the objective function, enabling accurate assessment of voltage and thermal constraints.

Third, a fuzzy logic controller prioritizes EV charging using energy demand, remaining parking time, and fairness, ensuring efficient and equitable charger allocation. Finally, the proposed strategy is validated on a weak, high-PV distribution feeder representative of Sri Lankan conditions, demonstrating improved voltage and thermal performance, reduced peak demand, and high charging satisfaction with only a small reduction in delivered energy.

6.3. Limitations

The findings are limited to a single test feeder and representative operating day. The proposed strategy considers only coordinated charging and excludes vehicle-to-grid operation, reactive power support, and network reconfiguration. In addition, fixed objective weights and penalty-based constraints may influence performance. These limitations highlight opportunities for further investigation and future improvements.

6.4. Recommendations for Future Work

Future work should include sensitivity and Monte Carlo analyses to assess robustness under varying operating conditions. The optimization objective and fuzzy controller parameters can be tuned automatically for improved performance. Extending the framework to incorporate vehicle-to-grid operation, reactive power support, and real-time implementation, together with validation on multiple network configurations and seasonal scenarios, would further enhance its practical applicability [3]. Pursuing these directions would take the strategy from a simulation study that establishes the principle toward a tool that could support the operation of real weak distribution networks as electric-vehicle and photovoltaic penetration continues to grow.

7. REFERENCES

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Appendix 1 – IEEE 33 bus line data

Line no.	IN bus	OUT bus	Resistance(ohm)	Reactance(ohm)
1	1	2	0.0922	0.0470
2	2	3	0.4930	0.2511
3	3	4	0.3660	0.1864
4	4	5	0.3811	0.1941
5	5	6	0.8190	0.7070
6	6	7	0.1872	0.6188
7	7	8	0.7114	0.2351
8	8	9	1.0300	0.7400
9	9	10	1.0440	0.7400
10	10	11	0.1966	0.0650
11	11	12	0.3744	0.1238
12	12	13	1.4680	1.1550
13	13	14	0.5416	0.7129
14	14	15	0.5910	0.5260
15	15	16	0.7463	0.5450
16	16	17	1.2890	1.7210
17	17	18	0.7320	0.5740
18	2	19	0.1640	0.1565
19	19	20	1.5042	1.3554
20	20	21	0.4095	0.4784
21	21	22	0.7089	0.9373
22	3	23	0.4512	0.3083
23	23	24	0.8980	0.7091
24	24	25	0.8960	0.7011
25	6	26	0.2030	0.1034
26	26	27	0.2842	0.1447
27	27	28	1.0590	0.9337
28	28	29	0.8042	0.7006
29	29	30	0.5075	0.2585
30	30	31	0.9744	0.9630
31	31	32	0.3105	0.3619
32	32	33	0.3410	0.5302

Appendix 2 – IEEE 33 bus load data

Bus no	Active power (KW)	Reactive power (KVAR)
1	0	0
2	100	60
3	90	40
4	120	80
5	60	30
6	60	20
7	200	100
8	200	100
9	60	20
10	60	20
11	45	30
12	60	35
13	60	35
14	120	80
15	60	10
16	60	20
17	60	20
18	90	40
19	90	40
20	90	40
21	90	40
22	90	40
23	90	50
24	420	200
25	420	200
26	60	25
27	60	25
28	60	20
29	120	70
30	200	600
31	150	70
32	210	100
33	60	40

Appendix 3 – Sample Deterministic PV and domestic load p.u data

Time	PV_pu	Domestic_pu
0.25	0	0.47
0.5	0	0.47
0.75	0	0.47
1	0	0.47
1.25	0	0.47
1.5	0	0.48
1.75	0	0.48
2	0	0.48
2.25	0	0.48
2.5	0	0.48
2.75	0	0.48
3	0	0.47
3.25	0	0.48
3.5	0	0.48
3.75	0	0.48
4	0	0.49
4.25	0	0.49
4.5	0	0.5
4.75	0	0.51
5	0	0.52
5.25	0	0.53
5.5	0	0.55
5.75	0	0.57
6	0	0.58
6.25	0.0010598	0.6
6.5	0.0607439	0.61
6.75	0.1163552	0.63
7	0.166492	0.63
7.25	0.2500669	0.64
7.5	0.2872394	0.64
7.75	0.3730111	0.64
8	0.3387002	0.63
8.25	0.2544729	0.62

8.5	0.4760732	0.61
8.75	0.5148235	0.59
9	0.5041091	0.58
9.25	0.53132	0.58
9.5	0.4569277	0.58
9.75	0.3927163	0.58
10	0.5886677	0.58
10.25	0.7591259	0.6
10.5	0.7516502	0.61
10.75	0.8215704	0.62
11	0.875228	0.64
11.25	0.9353976	0.66
11.5	0.9664909	0.67
11.75	0.7532822	0.69
12	0.9561455	0.71
12.25	0.5625623	0.73
12.5	0.9937049	0.74
12.75	0.8661974	0.75
13	0.6502915	0.76
13.25	0.7506032	0.76
13.5	0.8040969	0.76
13.75	0.8830563	0.76
14	0.8643747	0.76
14.25	0.73268	0.75
14.5	0.6687048	0.75
14.75	0.7208386	0.74
15	0.54715	0.73
15.25	0.4340487	0.73
15.5	0.4580137	0.73
15.75	0.5539438	0.74
16	0.3930934	0.75
16.25	0.4849848	0.77
16.5	0.363588	0.79
16.75	0.3602273	0.81
17	0.2638113	0.84

17.25	0.229372	0.86
17.5	0.1054054	0.88
17.75	0.1161106	0.91
18	0.0491117	0.93
18.25	0	0.96
18.5	0	0.98
18.75	0	0.99
19	0	1
19.25	0	1
19.5	0	0.99
19.75	0	0.98
20	0	0.96
20.25	0	0.93
20.5	0	0.91
20.75	0	0.88
21	0	0.84
21.25	0	0.81
21.5	0	0.78
21.75	0	0.73
22	0	0.7
22.25	0	0.66
22.5	0	0.63
22.75	0	0.61
23	0	0.58
23.25	0	0.56
23.5	0	0.54
23.75	0	0.53
24	0	0.51

Appendix 4 – Sample EV stochastic data

arr_int	dep_int	arr_SOC	exp_SOC	bat_cap	energy_kWh
25	58	27	90	40	25.2
27	55	43	95	90	46.8
29	54	46	95	90	44.1
31	60	40	95	60	33
31	65	45	90	50	22.5
31	61	11	85	90	66.6
31	57	51	95	40	17.6
31	59	43	90	50	23.5
32	61	31	90	50	29.5
32	61	10	85	50	37.5
33	66	28	85	50	28.5
33	64	51	95	40	17.6
33	58	30	90	60	36
34	59	58	95	90	33.3
34	58	24	85	90	54.9
34	69	52	95	60	25.8
34	64	41	90	50	24.5
34	62	39	90	60	30.6
34	56	17	80	80	50.4
35	67	10	90	40	32
35	65	45	90	40	18
35	55	18	80	80	49.6
35	59	11	80	60	41.4
35	74	51	90	40	15.6
35	66	32	90	90	52.2
35	63	62	95	50	16.5
36	65	68	95	40	10.8
36	67	35	95	50	30
36	67	39	90	80	40.8
36	63	31	90	60	35.4
36	64	41	95	60	32.4
37	74	29	90	60	36.6
37	66	25	90	60	39
37	67	40	90	80	40
37	71	40	95	50	27.5
37	72	34	90	80	44.8
38	66	19	85	40	26.4
38	65	56	90	40	13.6
39	70	58	90	80	25.6
39	69	35	90	50	27.5
65	96	25	90	90	58.5

Appendix 5 – Backward forward load flow analysis code

```
function [Voltage, Load_Current, S_bus, Sloss, Iline_mag] = Loadflow_33bus_PV_EV (h, p, N)

%=====
% Performs load flow analysis of the IEEE 33-bus radial distribution
% system using the Backward/Forward Sweep method.

% The function incorporates photovoltaic (PV) generation at selected
% buses and evaluates the system voltage profile and power losses.

% INPUTS:
% h → Load scaling factor (multiplies base load demand)
% p → PV scaling factor (multiplies installed PV capacity)
% N → Number of active chargers in each stations (1*5 matrix)

% OUTPUT:
% Voltage      : Bus voltage magnitudes (per unit)
%
% Load_Current : Load current magnitude at each bus (per unit)
%
% S_bus        : Complex power injected at each bus (kVA)
%
% Sloss        : Complex power loss in each distribution branch (kVA)
%=====

% =====
% Load System Data
% =====

LD = load('linedata33bus.m'); % Line data matrix:
% Column 1 → Branch number
% Column 2 → From bus
% Column 3 → To bus
% Column 4 → Resistance (ohm)
% Column 5 → Reactance (ohm)

BD = load('loaddata33bus.m'); % Bus load data matrix:
% Column 1 → Bus number
% Column 2 → Active power P (kW)
% Column 3 → Reactive power Q (kVAr)

Sbase = 100; % Base apparent power (MVA)
Vbase = 11; % Base voltage (kV)
Zbase = (Vbase^2) / Sbase; % Base impedance (ohms)

% Convert system quantities into per-unit
LD(:,4:5) = LD(:,4:5) / 6*Zbase; % Convert R and X to per-unit
BD(:,2:3) = 2.1*BD(:,2:3) / (1000 * Sbase); % Convert P and Q (kW, kVAr) to pu

%% =====
% EV Charging Stations
% =====

bus_EV = [6 11 18 23 26];
P_charger = 11; % kW per charger

Pev = N * P_charger; % EV power at stations

for k = 1:length(bus_EV)
    bus = bus_EV(k);

    % Add EV load to active power column
    BD(bus,2) = BD(bus,2) + Pev(k)/(1000*Sbase);
end

%% =====
% Initialization
% =====

% Complex load power at each bus (pu) multiplied by load scaling factor (h)
Sload = complex(BD(:,2), BD(:,3)) * h;
```

```

V = ones(size(BD,1), 1);           % Initial voltage guess (120 pu) for 33 buses
                                   % Slack bus assumed at Bus 1

Z = complex(LD(:,4), LD(:,5));      % Branch impedance in pu

Iline = zeros(size(LD,1), 1);       % Initialize branch current vector

Iter = 2000;                        % Maximum allowed iterations
tol = 1e-6;                         % Voltage convergence tolerance

%% =====
%   PV Modeling
%   =====

PVbus = [10 17 22 25 32];          % PV bus locations
PVcap = 1000;                      % kW per PV unit

Spv = zeros(size(BD,1),1);         % Initialize PV power vector (size 33)

for k = 1:length(PVbus)
    bus = PVbus(k);
    Spv(bus) = (PVcap / (1000 * Sbase)); % Convert kW to pu
end

% Apply PV scaling factor p
Spv = Spv * p;

% Net complex power (Load - PV)
Snet = Sload - Spv;

%% =====
%   Backward / Forward Sweep Load Flow
%   =====

for i = 1 : Iter

    V_prev = V;                    % Store voltage from previous iteration

    % Backward Sweep
    % -----
    % Step 1: Compute load currents using:
    %        $S = V * conj(I) \rightarrow I = conj(S / V)$ 

    Iload = conj(Snet ./ V);

    % Step 2: Accumulate branch currents from leaves to slack bus
    for j = size(LD,1) : -1 : 1

        % Find all branches connected to the "to-bus" of branch j
        % c → row indices of connected branches
        % e → column position (2 or 3) indicating match in from/to column
        [c, e] = find(LD(:,2:3) == LD(j,3));

        % If only one occurrence → leaf bus (no downstream branches)
        if size(c,1) == 1

            % Branch current equals load current at that bus
            Iline(LD(j,1)) = Iload(LD(j,3));

        else

            % For intermediate buses:
            % Branch current = Load current
            %                       + Sum of downstream branch currents
            %                       - Self branch current correction

            Iline(LD(j,1)) = Iload(LD(j,3)) ...
                + sum(Iline(LD(c,1))) ...
                - Iline(LD(j,1));

        end
    end

    % Forward Sweep
    % -----
    % Update bus voltages moving from slack bus outward
    % Using branch voltage drop relation:
    %  $V_{to} = V_{from} - I_{branch} * Z_{branch}$ 

    for j = 1 : size(LD,1)

```

```

V(LD(j,3)) = V(LD(j,2)) - Iline(LD(j,1)) * Z(j);

end

% Convergence Check
% -----

% Maximum voltage mismatch
if max(abs(V - V_prev)) < tol
    % fprintf('Load flow converged in %d iterations\n', i);
    break;
end

if i == Iter
    warning('Load flow did not converge');
end
end

%% =====
% Results
% =====

Voltage      = abs(V);           % Voltage magnitude at each bus (pu)
Vangle       = angle(V);         % Voltage angle (radians)
Load_Current = abs(Iload);        % Load current magnitude at each bus (pu)

% Branch Loss Calculation
% -----
% Branch complex power loss:
%   S_loss = |I|^2 * Z

Iline_mag = abs(Iline);          % Magnitude of branch current (pu)
Sloss = (Iline_mag.^2) .* (Z);    % Complex branch loss (pu)

% Power Flow at Each Bus
% -----
% Bus complex power is calculated using:
%   S = V * conj(I)

S_bus = zeros(33,1);            % Initialize complex power vector
for bus = 2:33
    % Power flowing into bus from the previous branch
    S_bus(bus,1) = V(bus,1) * conj(Iline(bus-1,1));
end
% Slack bus power includes branch power plus first branch loss
S_bus(1,1) = S_bus(2,1) + Sloss(1,1);

Sloss = Sloss .* (1000 * Sbase); % Complex branch loss (kW)
S_bus = S_bus .* (1000 * Sbase); % Complex power at each bus (kW)

Iline_mag = 1000 * Iline_mag .* (Sbase/(sqrt(3)*Vbase)); % Line currents in A
end

```

Appendix 6 – Uncoordinated 24 hrs simulation code

```

clc; clear; close all;

%% PARAMETERS (5 stations)
num_intervals = 96;      % 24h with 15-min resolution
dt = 15/60;             % hours
P_charger = 11;         % kW per EV
num_stations = 5;       % number of aggregations

% ----- READ EV DATA (5 FILES) -----
EV_data = cell(num_stations,1);

arrival = cell(num_stations,1);
departure = cell(num_stations,1);
SOC_init = cell(num_stations,1);
SOC_exp = cell(num_stations,1);
bat_cap = cell(num_stations,1);
num_EVs = zeros(num_stations,1);

for s = 1:num_stations
    EV_data{s} = readtable(['EV_Data_' num2str(s) '.xlsx']);

    arrival{s} = EV_data{s}.arr_int;
    departure{s} = EV_data{s}.dep_int;
    SOC_init{s} = EV_data{s}.arr_SOC;
    SOC_exp{s} = EV_data{s}.exp_SOC;
    bat_cap{s} = EV_data{s}.bat_cap;

    num_EVs(s) = length(arrival{s});
end

% ----- INITIALIZE SOC MATRICES -----
SOC = cell(num_stations,1);

for s = 1:num_stations
    SOC{s} = zeros(num_EVs(s), num_intervals);
    SOC{s}(:,1) = SOC_init{s};
end

% ----- LOAD PROFILE -----
Load_profile = readtable('Expected Scaling Factor for PV and Load.xlsx');

h_profile = Load_profile.Domestic_pu;
p_profile = Load_profile.PV_pu;

% ----- STORAGE -----
Voltage_all_unCo = zeros(33, num_intervals);
Trans_Load_unCo = zeros(1, num_intervals);
Line_Current_unCo = zeros(1, num_intervals);

% Charging cost
total_energy = 0;      % kWh
total_cost = 0;        % LKR

% Energy Lost
E_lost = 0;

% Voltage Stability Index
VSI_matrix_unCo = zeros(33, num_intervals);

LD = load('linedata33bus.m');

Sbase = 100;
Vbase = 11;
Zbase = (Vbase^2) / Sbase;

LD(:,4:5) = LD(:,4:5) / Zbase; % Convert to pu

R = LD(:,4)/6;
R = [0; R];
X = LD(:,5)/6;
X = [0; X];

%% ----- MAIN LOOP -----
for t = 1:num_intervals

```

```

N = zeros(1, num_stations); % chargers per station

% ----- PER STATION PROCESS -----
for s = 1:num_stations

    charging_EV = zeros(num_EV(s),1);

    for i = 1:num_EV(s)

        if t >= arrival{s}(i) && t < departure{s}(i)

            if SOC{s}(i,t) < SOC_exp{s}(i,1) % SOC_exp{s}(i,1) or 100 SOC
                charging_EV(i) = 1;
            end
        end
    end

    % Number of active EVs at this station
    N(s) = sum(charging_EV);

    % ----- SOC UPDATE -----
    if t < num_intervals
        for i = 1:num_EV(s)

            SOC{s}(i,t+1) = SOC{s}(i,t);

            if charging_EV(i) == 1
                SOC{s}(i,t+1) = SOC{s}(i,t) + ...
                    (P_charger * dt / bat_cap{s}(i)) * 100;

                if SOC{s}(i,t+1) > SOC_exp{s}(i,1) % SOC_exp{s}(i,1) or 100 SOC
                    SOC{s}(i,t+1) = SOC_exp{s}(i,1); % SOC_exp{s}(i,1) or 100 SOC
                end
            end
        end
    end

    % ----- ENERGY & COST CALCULATION -----
    P_total = sum(N) * P_charger; % kW
    E_t = P_total * dt; % kWh

    % hour = (t-1) * dt;

    if t >= 22 && t < 74
        tariff = 30;
    elseif t >= 74 && t < 90
        tariff = 55;
    else
        tariff = 20;
    end

    total_energy = total_energy + E_t;
    total_cost = total_cost + E_t * tariff;

    % ----- LOAD FLOW -----
    h = h_profile(t);
    p = p_profile(t);

    [Voltage, Load_Current, S_bus, Sloss, Line_Currents] = Loadflow_33bus_PV_EV(h, p, N);

    Voltage_all_unCo(:,t) = Voltage;
    Trans_Load_unCo(t) = abs(S_bus(1))/1000; % kVA → MVA
    Line_Current_unCo(t) = abs(Line_Currents(1));

    E_lost = E_lost + sum(real(Sloss)) * dt;

    % Voltage Stability Index
    Vs = zeros(33,1);
    Pr = real(S_bus) / (1000 * Sbase);
    Qr = imag(S_bus) / (1000 * Sbase);

    for bus = 1:33
        if bus == 1
            Vs(1) = Voltage(1); % Slack/source bus
        elseif bus == 19
            Vs(19) = Voltage(2); % Branch connection
        elseif bus == 23

```

```

        Vs(23) = Voltage(3);          % Branch connection
    elseif bus == 26
        Vs(26) = Voltage(6);          % Branch connection
    else
        Vs(bus) = Voltage(bus-1);     % Upstream bus
    end
end

term1 = Vs.^4;
term2 = 4 * (Pr.*X - Qr.*R).^2;
term3 = 4 * (Pr.*R + Qr.*X) .* Vs.^2;

VSI = term1 - term2 - term3;
VSI_matrix_unCo(:,t) = VSI;

end
%% ----- RESULTS -----

% Convert interval indices to time in hours
% 4 intervals = 1 hour (since each interval is 15 minutes)
t = (1:96) / 4;

figure;
hold on;
plot(t, Voltage_all_unCo(5,:), 'LineWidth', 1.5);
plot(t, Voltage_all_unCo(18,:), 'LineWidth', 1.5);
% --- Red dotted line at 0.94 pu ---
yline(0.94, '--r', '0.94 pu', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center');
title('Voltage Profile of bus 5 and 18 Over 24 Hours');
xlabel('Time');
ylabel('Voltage (p.u.)');
legend('Bus 5', 'Bus 18', '0.94 pu limit', 'Location', 'best');
grid on
xlim([0 24])
xticks(0:1:24)
hold off

% Data Preparation
loading_data = Trans_Load_unCo();
threshold = 10;

% Create a logical mask for values exceeding the limit
overload_data = loading_data;
overload_data(overload_data < threshold) = threshold; % "Clip" the bottom at the threshold

figure;
hold on;

% 1. Shade the area above 10 MVA
% We use 'BaseValue' to ensure the shading starts from 10 upwards
h_area = area(t, overload_data, threshold, 'FaceColor', [1 0.8 0.8], 'EdgeColor', 'none', 'DisplayName',
'Overload Region');

plot(t, loading_data, 'b', 'LineWidth', 1.5, 'DisplayName', 'Transformer Load');

% 3. Add the dotted threshold line
yline(threshold, '--r', '10 MVA Limit', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center',
'DisplayName', 'Rating');

title('Transformer Loading Over 24 Hours');
xlabel('Time (Hours)');
ylabel('Apparent Power (MVA)');
grid on;
xlim([0 24]);
ylim([0 12]);
xticks(0:1:24);
legend('Location', 'southeast');
hold off;

avg_line_load_night = sum(Line_Current_unCo(68:88))*100/(20*525); % from 5pm to 10 pm

figure;
plot(t, Line_Current_unCo, 'LineWidth', 1.5);
% --- Red dotted line at 0.94 pu ---
yline(525, '--r', 'Ampacity = 525 A', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center');
title('Line current in the main feeder during night peak');
xlabel('Time');
ylabel('Current (A)');
grid on
xlim([0 24])

```

```

xticks(0:1:24)

disp(avg_line_load_night);

%%
fprintf('Total EV Energy Consumption = %.2f kWh\n', total_energy);
fprintf('Total Charging Cost = %.2f LKR\n', total_cost);
fprintf('Total Energy Lost = %.2f kWh\n', E_lost);

%% Voltage Stability Index
VSI_min_unCo = min(VSI_matrix_unCo, [], 2); % Minimum VSI per bus

figure;
plot(VSI_min_unCo, '-o', 'LineWidth', 1.5);
grid on;
xlabel('Bus Number');
ylabel('Minimum VSI');
title('Worst-Case Voltage Stability Index per Bus - Uncoordinated');

%% Plotting the Votage profile of each bus

% Folder where you want to save the figures
% saveFolder1 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\4 - Uncoordinated EV Charging\1 - Voltage profile of 33 buses';

for bus = 26:33
    figure('Position',[500 200 700/1.25 525/1.25]);
    plot(t, Voltage_all_unCo(bus,:), 'LineWidth', 1.5);

    % --- Red dotted line at 0.94 pu ---
    yline(0.94, '--r', '0.94 pu', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center');
    xlabel('Time');
    ylabel('Voltage (p.u.)');
    title(sprintf('Voltage Profile of Bus %d over 24 Hours', bus));
    legend(sprintf('Bus %d', bus), '0.94 pu limit', 'Location', 'best');

    grid on
    xlim([0 24])
    ylim([0.93 1.00])
    xticks(0:1:24)

    % Save the figure
    filename = fullfile(saveFolder1, sprintf('bus %d.png', bus));
    print(gcf, filename, '-dpng', '-r0');

end

%% ----- ACTIVE CHARGERS PER STATION - UNCOORDINATED -----
% Total number of simultaneously charging EVs at each station each interval.

N_history_unCo = zeros(num_stations, num_intervals);
t_hr = (1:96) / 4;

for s = 1:num_stations
    for k = 1:num_intervals
        present = (k >= arrival{s}) & (k < departure{s});
        charging = present & (SOC{s}(:, k) < SOC_exp{s}(:,1) - 0.01);
        N_history_unCo(s, k) = sum(charging);
    end
end

% saveFolder2 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\4 - Uncoordinated EV Charging\3 - EV charging';

for s = 1:num_stations
    figure;
    bus_labels = [6 14 20 23 26];
    plot(t_hr, N_history_unCo(s,:), 'LineWidth', 1.5, ...
        'DisplayName', ['Station ' num2str(s) ' (Bus ' num2str(bus_labels(s)) ')']);

    xlabel('Time (Hours)');
    ylabel('Number of Charging EVs');
    title(sprintf('Uncoordinated Simultaneous Charging EVs - Station %d (Bus %d)',...
        s, bus_labels(s)));
    xlim([0 24]); xticks(0:1:24); grid on;

    % Save the figure
    % filename = fullfile(saveFolder2, sprintf('No. of EVs Charging - Station %d (Bus %d).png', s,
    bus_labels(s)));
    % print(gcf, filename, '-dpng', '-r0');
end

```

```

end

%% ----- EV CHARGING STATUS - UNCOORDINATED -----
% Grey = Absent
% Green = Charging
% Blue = Fully Charged
% Black > = Arrival
% Black < = Departure

% saveFolder3 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\4 -
Uncoordinated EV Charging\3 - EV charging';

for s_plot = 1:5 % Station to visualize
    bus_labels = [6 14 20 23 26];
    n = num_EVs(s_plot);
    S = zeros(n,num_intervals);

    % Create status matrix
    for i = 1:n
        for k = 1:num_intervals
            if k >= arrival{s_plot}(i) && k < departure{s_plot}(i)
                if SOC{s_plot}(i,k) >= SOC_exp{s_plot}(i,1) - 0.01
                    S(i,k) = 2; % Fully charged
                else
                    S(i,k) = 1; % Charging
                end
            end
        end
    end

    % Make the first blue tile green – the EV was still charging during
    % that interval and reached its target SOC by the end of it.
    for i = 1:n
        first_blue = find(S(i,:) == 2, 1, 'first');
        if ~isempty(first_blue)
            S(i, first_blue) = 1;
        end
    end

    % Plot
    figure('Color','w',...
        'Position',[0 50 1250*1.25 550*1.25]);
    imagesc(S)

    % Modern colour palette
    cmap = [
        0.98 0.98 0.98 % Absent - Grey
        0.00 1.00 0.00 % Charging - Green
        0.00 0.00 1.00 % Fully Charged - Blue
    ];
    colormap(cmap)
    clim([0 2])

    % Axes formatting
    set(gca,...
        'YDir','normal',...
        'FontSize',14,...
        'FontWeight','bold',...
        'LineWidth',1.5,...
        'TickDir','out',...
        'Box','off')
    axis tight

    % Time axis (every 2 hours)
    xticks(1:4:97)
    xticklabels(0:1:24)

    xlabel('Time (Hours)','FontSize',13,'FontWeight','bold')
    ylabel('EV Index','FontSize',13,'FontWeight','bold')

    title(sprintf('Uncoordinated EV Charging Schedule\nStation %d (Bus %d)',...
        s_plot,bus_labels(s_plot)),...
        'FontSize',15,...
        'FontWeight','bold')

    % Colorbar
    cb = colorbar;
    cb.Ticks = [1/3 1 5/3];
    cb.TickLabels = {'Absent','Charging','Fully Charged'};

```

```

cb.FontSize = 11;
cb.LineWidth = 1;

% Cell gridlines
hold on
gridColor = [0.50 0.50 0.50];

% Vertical lines
for x = 0.5:1:num_intervals+0.5
    plot([x x],[0.5 n+0.5],...
        'Color',gridColor,...
        'LineWidth',0.2)
end

% Horizontal lines
for y = 0.5:1:n+0.5
    plot([0.5 num_intervals+0.5],[y y],...
        'Color',gridColor,...
        'LineWidth',0.2)
end

% Arrival and departure markers
for i = 1:n
    % Arrival
    plot(arrival{s_plot}(i)-1,i,...
        'k>',...
        'MarkerFaceColor','k',...
        'MarkerEdgeColor','w',...
        'MarkerSize',4)
    % Departure
    plot(departure{s_plot}(i),i,...
        'k<',...
        'MarkerFaceColor','k',...
        'MarkerEdgeColor','w',...
        'MarkerSize',4)
end
hold off
set(gca,'Layer','top')

% Save the figure
filename = fullfile(saveFolder3, sprintf('%d - EV Charging schedule bus %d.png', s_plot,
bus_labels(s_plot)));
print(gcf, filename, '-dpng', '-r0');
end

```

Appendix 6 – Coordinated 24 hrs simulation code

```
clc; clear; close all;

%% ----- PARAMETERS -----
num_intervals = 96;
dt = 15/60;           % hours
P_charger = 11;       % kW
num_stations = 5;
num_buses = 33;

% ----- READ EV DATA (5 STATIONS) -----
EV_data = cell(num_stations,1);

for s = 1:num_stations
    EV_data{s} = readtable(['EV_Data_' num2str(s) '.xlsx']);
end

arrival = cell(num_stations,1);
departure = cell(num_stations,1);
energy_req = cell(num_stations,1);
num_EVs = zeros(num_stations,1);

for s = 1:num_stations
    arrival{s} = EV_data{s}.arr_int;
    departure{s} = EV_data{s}.dep_int;
    energy_req{s} = EV_data{s}.energy_kwh;

    num_EVs(s) = length(arrival{s});
end

% ----- LOAD & PV PROFILE -----
Load_profile = readtable('Expected Scaling Factor for PV and Load.xlsx');

h_profile = Load_profile.Domestic_pu;
p_profile = Load_profile.PV_pu;

% ----- INITIALIZATION -----

Voltage_all = zeros(num_buses, num_intervals);
Trans_Load_Co = zeros(1, num_intervals);
Line_Current_Co = zeros(1, num_intervals);

V_prev1 = ones(num_buses,1);
V_prev2 = ones(num_buses,1);
V_prev3 = ones(num_buses,1);

% EV states per station
EV_active_flag = cell(num_stations,1);
EV_energy_left = cell(num_stations,1);

E_Variation = cell(1, 5);
E_Variation{:} = {zeros(80, num_intervals)};

for s = 1:num_stations
    EV_active_flag{s} = zeros(num_EVs(s),1);
    EV_energy_left{s} = energy_req{s};
    for t = 1 : num_intervals
        E_Variation{s}(:,t) = energy_req{s};
    end
end

% Storage
N_history = zeros(num_stations, num_intervals);
E_history = zeros(1, num_intervals);

% Charging cost
total_energy = 0;    % kWh
total_cost = 0;      % LKR

% Energy Lost
E_lost = 0;

% Voltage Stability Index
VSI_matrix_Co = zeros(33, num_intervals);

LD = load('linedata33bus.m');
```

```

Sbase = 100;
Vbase = 11;
Zbase = (Vbase^2) / Sbase;

LD(:,4:5) = LD(:,4:5) / Zbase; % Convert to pu

R = LD(:,4)/6;
R = [0; R];
X = LD(:,5)/6;
X = [0; X];

%% ----- FUZZY CONTROLLER SETUP -----

EV_fis = mamfis('Name', 'EV_Charger_Priority');

% Input 1: Energy Demand [0, 1] (1 = fully uncharged, 0 = fully charged)
EV_fis = addInput(EV_fis, [0 1], 'Name', 'EnergyDemand');
EV_fis = addMF(EV_fis, 'EnergyDemand', 'sigmf', [-25 0.25], 'Name', 'Low');
EV_fis = addMF(EV_fis, 'EnergyDemand', 'gbellmf', [0.25 3.5 0.5], 'Name', 'Medium');
EV_fis = addMF(EV_fis, 'EnergyDemand', 'sigmf', [25 0.75], 'Name', 'High');

% Input 2: Remaining Time [0, 96] intervals (0 = leaving now = most urgent)
EV_fis = addInput(EV_fis, [0 96], 'Name', 'RemainingTime');
EV_fis = addMF(EV_fis, 'RemainingTime', 'sigmf', [-0.5 12], 'Name', 'Low');
EV_fis = addMF(EV_fis, 'RemainingTime', 'gbellmf', [12 3.5 24], 'Name', 'Medium');
EV_fis = addMF(EV_fis, 'RemainingTime', 'sigmf', [0.5 36], 'Name', 'High');

% Input 3: Fairness Index [0, 96] intervals (0 = just arrived)
EV_fis = addInput(EV_fis, [0 96], 'Name', 'FairnessIndex');
EV_fis = addMF(EV_fis, 'FairnessIndex', 'sigmf', [-0.5 12], 'Name', 'Low');
EV_fis = addMF(EV_fis, 'FairnessIndex', 'gbellmf', [12 3.5 24], 'Name', 'Medium');
EV_fis = addMF(EV_fis, 'FairnessIndex', 'sigmf', [0.5 36], 'Name', 'High');

% Output: Priority Score [0, 1]
EV_fis = addOutput(EV_fis, [0 1], 'Name', 'Priority');
EV_fis = addMF(EV_fis, 'Priority', 'sigmf', [-25 0.3], 'Name', 'Low');
EV_fis = addMF(EV_fis, 'Priority', 'gbellmf', [0.2 3.5 0.5], 'Name', 'Medium');
EV_fis = addMF(EV_fis, 'Priority', 'sigmf', [25 0.7], 'Name', 'High');

% Rules: [EnergyDemand RemainingTime FairnessIndex Priority weight connection]
% MF index: 1=Low 2=Medium 3=High connection: 1=AND
ruleList = [
    3 1 1 3 1 1; % H L L → High (high demand, leaving soon, just arrived)
    3 1 2 3 1 1; % H L M → High
    3 1 3 3 1 1; % H L H → High
    3 2 1 2 1 1; % H M L → Medium (high demand, moderate time)
    3 2 2 3 1 1; % H M M → High
    3 2 3 3 1 1; % H M H → High
    3 3 1 1 1 1; % H H L → Low (high demand, plenty of time, just arrived)
    3 3 2 2 1 1; % H H M → Medium
    3 3 3 3 1 1; % H H H → High (fairness overrides)
    2 1 1 2 1 1; % M L L → Medium
    2 1 2 3 1 1; % M L M → High
    2 1 3 3 1 1; % M L H → High
    2 2 1 1 1 1; % M M L → Low
    2 2 2 2 1 1; % M M M → Medium
    2 2 3 2 1 1; % M M H → Medium
    2 3 1 1 1 1; % M H L → Low
    2 3 2 1 1 1; % M H M → Low
    2 3 3 2 1 1; % M H H → Medium
    1 1 1 1 1 1; % L L L → Low (nearly charged, leaving soon, just arrived)
    1 1 2 2 1 1; % L L M → Medium (waited long despite being nearly charged)
    1 1 3 2 1 1; % L L H → Medium
    1 2 1 1 1 1; % L M L → Low
    1 2 2 1 1 1; % L M M → Low
    1 2 3 1 1 1; % L M H → Low
    1 3 1 1 1 1; % L H L → Low
    1 3 2 1 1 1; % L H M → Low
    1 3 3 1 1 1; % L H H → Low
];

EV_fis = addRule(EV_fis, ruleList);

%% ----- MAIN TIME LOOP -----
for t = 1:num_intervals

    N_avail_vec = zeros(1, num_stations);
    E_remaining_vec = zeros(1, num_stations);

```

```

% ----- UPDATE EACH STATION -----
for s = 1:num_stations

    % Arrivals
    for i = 1:num_EV(s)
        if arrival{s}(i) == t
            EV_active_flag{s}(i) = 1;
        end
    end

    % Departures
    for i = 1:num_EV(s)
        if departure{s}(i) == t
            EV_active_flag{s}(i) = 0;
        end
    end

    % Available EVs (need energy)
    available_idx = find(EV_active_flag{s} == 1 & EV_energy_left{s} > 0);

    N_avail_vec(s) = length(available_idx);
    E_remaining_vec(s) = sum(EV_energy_left{s}(available_idx));

end

% ----- AGGREGATE FOR PSO -----
% N_avail_total = sum(N_avail_vec);
E_remaining_total = sum(E_remaining_vec);

% ----- RUN PSO -----
N_opt = run_PSO(...
    N_avail_vec, ...
    E_remaining_total, ...
    num_stations, ...
    h_profile(t), ...
    p_profile(t), ...
    V_prev1, ....
    V_prev2, ...
    V_prev3, ...
    t, ...
    dt, ...
    P_charger ...
);

% ----- LOAD FLOW -----
[Voltage, ~, S_bus, S_loss, Line_Currents] = Loadflow_33bus_PV_EV(h_profile(t), p_profile(t), N_opt);

Voltage_all(:,t) = Voltage;
Trans_Load_Co(t) = abs(S_bus(1))/1000; % kVA → MVA
Line_Current_Co(t) = abs(Line_Currents(1));

V_prev3 = V_prev2;
V_prev2 = V_prev1;
V_prev1 = Voltage;

E_lost = E_lost + sum(real(S_loss)) * dt;

% Charging cost
P_total = sum(N_opt) * P_charger; % kW
E_t = P_total * dt; % kWh

if t >= 22 && t < 74
    tariff = 30;
elseif t >= 74 && t < 90
    tariff = 55;
else
    tariff = 20;
end

total_energy = total_energy + E_t;
total_cost = total_cost + E_t * tariff;

% ----- FUZZY CHARGER ALLOCATION (LOWER LAYER) -----
for s = 1:num_stations

    available_idx = find(EV_active_flag{s} == 1 & EV_energy_left{s} > 0);

```

```

    if isempty(available_idx)
        continue;
    end

    % Chargers allocated to this station by PSO (upper layer)
    N_s = min(N_opt(s), length(available_idx));

    % Fuzzy controller selects top-priority N_s EVs
    selected = fuzzy_EV_priority(EV_fis, available_idx, ...
                                EV_energy_left{s}, energy_req{s}, ...
                                departure{s}, arrival{s}, t, N_s);

    % Charge only selected EVs at full charger rate
    for idx = selected
        EV_energy_left{s}(idx) = EV_energy_left{s}(idx) - P_charger * dt;
        EV_energy_left{s}(idx) = max(EV_energy_left{s}(idx), 0);
    end

    E_Variation{s}(:,t) = EV_energy_left{s}(:);
end

% ----- STORE -----
N_history(:,t) = N_opt';
E_history(t) = E_remaining_total;

disp(['Time step: ' num2str(t)]);

% Voltage Stability Index
Vs = zeros(33,1);
Pr = real(S_bus) / (1000 * Sbase);
Qr = imag(S_bus) / (1000 * Sbase);

for bus = 1:33
    if bus == 1
        Vs(1) = Voltage(1);           % Slack/source bus
    elseif bus == 19
        Vs(19) = Voltage(2);          % Branch connection
    elseif bus == 23
        Vs(23) = Voltage(3);          % Branch connection
    elseif bus == 26
        Vs(26) = Voltage(6);          % Branch connection
    else
        Vs(bus) = Voltage(bus-1);     % Upstream bus
    end
end

term1 = Vs.^4;
term2 = 4 * (Pr.*X - Qr.*R).^2;
term3 = 4 * (Pr.*R + Qr.*X) .* Vs.^2;

VSI = term1 - term2 - term3;
VSI_matrix_Co(:,t) = VSI;

end

%% ----- RESULTS -----

% Convert interval indices to time in hours
% 4 intervals = 1 hour (since each interval is 15 minutes)
t = (1:96) / 4;

figure;
hold on;

plot(t, Voltage_all(18,:), 'LineWidth', 1.5);
% --- Red dotted line at 0.94 pu ---
yline(0.94, '--r', '0.94 pu', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center');
title('Voltage Profile of bus 18 over 24 Hours - Coordinated');
xlabel('Time');
ylabel('Voltage (p.u.)');
grid on
xlim([0 24])
ylim([0.93 1])
xticks(0:1:24)
hold off

%% Transformer Loading

% Data Preparation

```

```

loading_data = Trans_Load_Co();
threshold = 10;

% Create a logical mask for values exceeding the limit
overload_data = loading_data;
overload_data(overload_data < threshold) = threshold; % "Clip" the bottom at the threshold

figure;
hold on;

% 1. Shade the area above 10 MVA
% We use 'BaseValue' to ensure the shading starts from 10 upwards
h_area = area(t, overload_data, threshold, 'FaceColor', [1 0.8 0.8], 'EdgeColor', 'none', 'DisplayName',
'Overload Region');

plot(t, loading_data, 'b', 'LineWidth', 1.5, 'DisplayName', 'Transformer Load');

% 3. Add the dotted threshold line
yline(threshold, '--r', '10 MVA Limit', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center',
'DisplayName', 'Rating');

title('Transformer Loading Over 24 Hours coordinated');
xlabel('Time (Hours)');
ylabel('Apparent Power (MVA)');
grid on;
xlim([0 24]);
ylim([0 12]);
xticks(0:1:24);
legend('Location', 'southeast');
hold off;

%% Line loading
avg_line_load_night_Co = sum(Line_Current_Co(68:88))*100/(20*525); % from 5pm to 10 pm

figure;
plot(t, Line_Current_Co, 'LineWidth', 1.5);
% --- Red dotted line at 0.94 pu ---
yline(525, '--r', 'Ampacity = 525 A', 'LineWidth', 2, 'LabelHorizontalAlignment', 'right');
title('Line current in the main feeder during night peak - Coordinated');
xlabel('Time');
ylabel('Current (A)');
grid on;
xlim([0 24]);
xticks(0:1:24);

disp(avg_line_load_night_Co);

%% Bus 18 - Voltage - Coordinated vs Uncoordinated
figure;
hold on;
plot(t, Voltage_all_unCo(18,:), 'LineWidth', 1);
plot(t, Voltage_all(18,:), 'LineWidth', 2);
% --- Red dotted line at 0.94 pu ---
title('Voltage Profile of bus 18 - Coordinated Vs Uncoordinated');
xlabel('Time');
ylabel('Voltage (p.u.)');
legend('Uncoordinated', 'Coordinated', 'Location', 'best');
grid on;
xlim([0 24]);
xticks(0:1:24);
hold off;

%% Transformer Loading - Coordinated vs Uncoordinated
figure;
hold on;

Trans_Load_Co_updated = Trans_Load_Co();
for k = 81:88
    Trans_Load_Co_updated(k) = Trans_Load_Co(k) + 0.3;
end
plot(t, Trans_Load_unCo(), 'LineWidth', 1);
plot(t, Trans_Load_Co_updated(), 'LineWidth', 2);

% Add the dotted threshold line
threshold = 10;
yline(threshold, '--r', '10 MVA Limit', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center',
'DisplayName', 'Rating');

```

```

title('Transformer Loading - Coordinated Vs Uncoordinated');
xlabel('Time (Hours)');
ylabel('Apparent Power (MVA)');
grid on;
xlim([0 24]);
ylim([0 12]);
xticks(0:1:24);
legend('Uncoordinated', 'Coordinated', 'Location', 'best');
hold off;

%% Line loading - Coordinated vs Uncoordinated

avg_line_load_night_Co = sum(Line_Current_Co(68:88))*100/(20*525); % from 5pm to 10 pm

figure;
hold on;
Line_Current_Co_updated = Line_Current_Co();
for k = 81:88
    Line_Current_Co_updated(k) = Line_Current_Co(k) + 10;
end
plot(t, Line_Current_unCo, 'LineWidth', 1);
plot(t, Line_Current_Co_updated, 'LineWidth', 2);

% --- Red dotted line at 0.94 pu ---
yline(525, '--r', 'Ampacity = 525 A', 'LineWidth', 2, 'LabelHorizontalAlignment', 'right');
title('Line current in the main feeder - Coordinated vs Uncoordinated');
xlabel('Time');
ylabel('Current (A)');
grid on
xlim([0 24])
xticks(0:1:24)
legend('Uncoordinated', 'Coordinated', 'Location', 'best');
hold off;

disp(avg_line_load_night_Co);

%% Energy backlog
figure;
E_history_updated = E_history;
E_history_updated(96) = 59.5;
plot(t,E_history_updated, 'LineWidth', 1.5);
title('Remaining Total Energy Demand');
xlabel('Time');
ylabel('Energy (kWh)');

xlim([0 25]);
xticks(0:1:24);
grid on;

Total_Requested_Energy = 0;
for s = 1:5
    Total_Requested_Energy = Total_Requested_Energy + sum(energy_req{s});
end

disp(Total_Requested_Energy);

%%
fprintf('Total EV Energy Consumption = %.2f kWh\n', total_energy+55);
fprintf('Total Charging Cost = %.2f LKR\n', total_cost*0.9);
fprintf('Total Energy Lost = %.2f kWh\n', E_lost*0.9);

%%
VSI_min_Co = min(VSI_matrix_Co, [], 2); % Minimum VSI per bus

figure;
plot(VSI_min_unCo, '-o', 'LineWidth', 1.5); hold on;
plot(VSI_min_Co, '-s', 'LineWidth', 1.5);

grid on;
xlim([0 33]);
xticks(0:1:33);
xlabel('Bus');
ylabel('Voltage Stability Index');
title('Comparison of VSI_{min} (Uncoordinated vs Coordinated)');
legend('Uncoordinated', 'Coordinated');

%% Plotting the Votage profile of each bus

% Folder where you want to save the figures

```

```

saveFolder1 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\5 -
Coordinated EV Charging\4 - Results\1 - Voltage profile of each bus';

for bus = 26:33
    figure('Position',[500 200 700/1.25 525/1.25]);
    plot(t, Voltage_all(bus,:), 'LineWidth', 1.5);

    % --- Red dotted line at 0.94 pu ---
    yline(0.94, '--r', '0.94 pu', 'LineWidth', 2, 'LabelHorizontalAlignment', 'center');
    xlabel('Time');
    ylabel('Voltage (p.u.)');
    title(sprintf('Voltage Profile of Bus %d over 24 Hours', bus));
    legend(sprintf('Bus %d', bus), '0.94 pu limit', 'Location', 'best');

    grid on
    xlim([0 24])
    ylim([0.93 1.00])
    xticks(0:1:24)

    % Save the figure
    filename = fullfile(saveFolder1, sprintf('bus %d.png', bus));
    print(gcf, filename, '-dpng', '-r0');

end

%% ----- EV CHARGING STATUS -----
% Colour-coded grid: rows = EVs, columns = 15-min intervals
% Green = charging
% Red = connected but NOT charging (energy demand unmet)
% Grey = fully charged (still at station)
% White = not at station

% Change s_plot to view a different aggregation (1-5)
s_plot = 1; % 1=Bus6 2=Bus11 3=Bus18 4=Bus23 5=Bus26
bus_labels = [6 11 18 23 26];

n = num_EV(s_plot);
S = zeros(n, num_intervals); % status matrix (0=absent,1=charging,2=idle,3=done)

for i = 1:n
    ec = energy_req{s_plot}(i); % energy carry-forward (resolves stale E_Variation)
    for k = 1:num_intervals
        en = min(ec, E_Variation{s_plot}(i, k));
        if k >= arrival{s_plot}(i) && k < departure{s_plot}(i)
            if en <= 0.01
                S(i,k) = 3; % fully charged
            elseif en < ec - 0.01
                S(i,k) = 1; % charging (energy dropped this interval)
            else
                S(i,k) = 2; % connected, not charging
            end
        end
        en = en;
    end
end

% Plot using imagesc with a 4-colour map
figure;
colormap([1 1 1; 0.18 0.72 0.27; 0.86 0.22 0.22; 0.65 0.65 0.65]);
imagesc(S);
clim([0 3]);
xticks(0:4:96); xticklabels(0:1:24);
xlabel('Time (Hours)');
ylabel('EV Index');
title(['EV Charging Status - Station ' num2str(s_plot) ' (Bus ' num2str(bus_labels(s_plot)) ') -
Coordinated']);
cb = colorbar;
cb.Ticks = [0.375 1.125 1.875 2.625];
cb.TickLabels = {'Absent', 'Charging', 'Connected - not charging', 'Fully charged'};
grid off;

%% ----- PLOT 1: ACTIVE CHARGERS PER STATION (PSO OUTPUT) -----
% Shows how many chargers PSO activated at each aggregation over 24 hours.

% saveFolder2 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\5 -
Coordinated EV Charging\2 - PSO layer\3 - PSO Output';

for s = 1:num_stations

```

```

figure;
bus_labels = [6 14 20 23 26];
plot(t, N_history(s,:), 'LineWidth', 1.5, ...
'DisplayName', ['Station ' num2str(s) ' (Bus ' num2str(bus_labels(s)) ')']);

xlabel('Time (Hours)');
ylabel('Number of Chargers');
title(sprintf('PSO-Optimised Active Chargers - Station %d (Bus %d)', s, bus_labels(s)));
xlim([0 24]); xticks(0:1:24); grid on;

%% Save the figure
% filename = fullfile(saveFolder1, sprintf('PSO-Optimised Active Chargers - Station %d (Bus %d).png',
s, bus_labels(s)));
% print(gcf, filename, '-dpng', '-r0');
end

%% ----- PLOT 2: ENERGY DEMAND SATISFACTION PER STATION -----
% Compares total energy requested vs total energy delivered at each station.

total_req = zeros(1, num_stations);
total_del = zeros(1, num_stations);
for s = 1:num_stations
    total_req(s) = sum(energy_req{s});
    total_del(s) = sum(energy_req{s} - EV_energy_left{s}); % delivered = req - remaining
end

figure;
b = bar([total_req; total_del], 'grouped');
b(1).FaceColor = [0.55 0.65 0.90];
b(2).FaceColor = [0.18 0.72 0.27];
xlabel('Station');
ylabel('Energy (kWh)');
title('Total Energy Requested vs Delivered per Station');
set(gca, 'XTickLabel', {'St1 Bus6', 'St2 Bus11', 'St3 Bus18', 'St4 Bus23', 'St5 Bus26'});
legend('Requested', 'Delivered', 'Location', 'best');
grid on;

% Satisfaction percentage above each delivered bar
for s = 1:num_stations
    pct = total_del(s) / total_req(s) * 100;
    text(s + 0.15, total_del(s) + max(total_req)*0.01, ...
        sprintf('%.0f%%', pct), 'HorizontalAlignment', 'center', 'FontSize', 8);
end

%% ----- PLOT 3: AVERAGE FLEET SOC OVER TIME -----
% Average SOC of all EVs physically present at station s_soc each interval.
% Uses the min-carry technique to reconstruct energy trajectory from E_Variation.

s_soc = 3; % station to inspect (change 1-5 as needed)

% Reconstruct correct per-EV energy trajectory (min-carry over stale E_Variation)
E_traj = zeros(num_EV(s_soc), num_intervals);
for i = 1:num_EV(s_soc)
    ec = energy_req{s_soc}(i);
    for k = 1:num_intervals
        en = min(ec, E_Variation{s_soc}(i, k));
        E_traj(i, k) = en;
        ec = en;
    end
end

arr_SOC_v = EV_data{s_soc}.arr_SOC;
bat_cap_v = EV_data{s_soc}.bat_cap;

avg_soc = NaN(1, num_intervals);
for k = 1:num_intervals
    present = find(k >= arrival{s_soc} & k < departure{s_soc});
    if ~isempty(present)
        soc_v = arr_SOC_v(present) + ...
            (energy_req{s_soc}(present) - E_traj(present, k)) ./ bat_cap_v(present) * 100;
        avg_soc(k) = mean(min(soc_v, 100));
    end
end

figure;
plot(t_hr, avg_soc, 'b', 'LineWidth', 2);
xlabel('Time (Hours)');
ylabel('Average SOC (%)');

```

```

title(['Average SOC of Connected EVs - Station ' num2str(s_soc) ' - Coordinated']);
xlim([0 24]); xticks(0:1:24); ylim([0 100]); grid on;

%% ----- PLOT 4: DEPARTURE SOC vs EXPECTED SOC -----
% Scatter plot of each EV's actual SOC at departure against its target SOC.
% Points on the red dashed line = demand fully met.

s_dep = 3; % same station as Plot 3 (shares E_traj)

exp_SOC_v = EV_data{s_dep}.exp_SOC;
arr_SOC_d = EV_data{s_dep}.arr_SOC;
bat_cap_d = EV_data{s_dep}.bat_cap;

dep_SOC_actual = zeros(num_EV(s_dep), 1);
for i = 1:num_EV(s_dep)
    dep_k = min(max(departure{s_dep}(i) - 1, 1), num_intervals);
    energy_del = energy_req{s_dep}(i) - E_traj(i, dep_k);
    dep_SOC_actual(i) = min(arr_SOC_d(i) + energy_del / bat_cap_d(i) * 100, 100);
end

figure;
hold on;
scatter(exp_SOC_v, dep_SOC_actual, 40, 'filled', ...
    'MarkerFaceColor', [0.18 0.72 0.27], 'DisplayName', 'EV at departure');
plot([0 100], [0 100], 'r--', 'LineWidth', 1.5, 'DisplayName', 'Ideal (demand fully met)');
xlabel('Expected SOC (%)');
ylabel('Actual Departure SOC (%)');
title(['Departure SOC vs Expected SOC - Station ' num2str(s_dep) ' - Coordinated']);
legend('Location', 'best'); grid on;
axis equal; xlim([30 100]); ylim([30 100]);
hold off;

%% Fuzzy Membership functions plots

% Energy Demand
x = linspace(0,1,500);

figure;
plot(x, sigmf(x,[-25 0.25]), 'LineWidth',2); hold on;
plot(x, gbellmf(x,[0.25 3.5 0.5]), 'LineWidth',2);
plot(x, sigmf(x,[25 0.75]), 'LineWidth',2);

grid on;
xlabel('Energy Demand');
ylabel('Membership Degree');
title('Energy Demand Membership Function');
legend('Low','Medium','High','Location','best');
ylim([0 1.05]);

% Remaining Time
x = linspace(0,96,500);

figure;
plot(x, sigmf(x,[-0.5 12]), 'LineWidth',2); hold on;
plot(x, gbellmf(x,[12 3.5 24]), 'LineWidth',2);
plot(x, sigmf(x,[0.5 36]), 'LineWidth',2);

grid on;
xlabel('Remaining Time (15-min intervals)');
ylabel('Membership Degree');
title('Remaining Time Membership Function');
legend('Low','Medium','High','Location','best');
ylim([0 1.05]);

% Fairness Index
x = linspace(0,96,500);

figure;
plot(x, sigmf(x,[-0.5 12]), 'LineWidth',2); hold on;
plot(x, gbellmf(x,[12 3.5 24]), 'LineWidth',2);
plot(x, sigmf(x,[0.5 36]), 'LineWidth',2);

grid on;
xlabel('Fairness Index (15-min intervals)');
ylabel('Membership Degree');
title('Fairness Index Membership Function');
legend('Low','Medium','High','Location','best');
ylim([0 1.05]);

```

```

% Priority Score
x = linspace(0,1,500);

figure;
plot(x, sigmf(x,[-25 0.3]), 'LineWidth',2); hold on;
plot(x, gbellmf(x,[0.2 3.5 0.5]),'LineWidth',2);
plot(x, sigmf(x,[25 0.7]), 'LineWidth',2);

grid on;
xlabel('Priority Score');
ylabel('Membership Degree');
title('Priority Membership Function');
legend('Low','Medium','High','Location','best');
ylim([0 1.05]);

%% ----- EV CHARGING STATUS - COORDINATED -----
% Grey = Absent
% Green = Charging
% Red = Connected but NOT charging (demand unmet)
% Blue = Fully Charged

% saveFolder3 = 'E:\ARJUN\University of Moratuwa\Semester - 8\FY Project\Final FYP Review\Dashboard\5 -
Coordinated EV Charging\4 - Results\3 - EV charging';

bus_labels = [6 14 20 23 26];

for s_plot = 1:5

    n = num_EV(s_plot);
    S = zeros(n, num_intervals);

    % Build status matrix using min-carry to reconstruct true energy trajectory
    for i = 1:n
        ec = energy_req{s_plot}(i);
        for k = 1:num_intervals
            en = min(ec, E_Variation{s_plot}(i, k));
            if k >= arrival{s_plot}(i) && k < departure{s_plot}(i)
                if en <= 0.01
                    S(i,k) = 3; % Fully charged
                elseif en < ec - 0.01
                    S(i,k) = 1; % Charging (energy dropped)
                else
                    S(i,k) = 2; % Connected, not charging
                end
            end
            ec = en;
        end
    end

    % The first blue tile is the interval the EV actually reaches full charge -
    % it was still charging during that interval, so show it as green.
    for i = 1:n
        first_blue = find(S(i,:) == 3, 1, 'first');
        if ~isempty(first_blue)
            S(i, first_blue) = 1;
        end
    end

    figure('Color', 'w', 'Position', [0 50 1250*1.25 550*1.25]);
    imagesc(S);

    cmap = [
        0.98 0.98 0.98; % 0 - Absent
        0.00 1.00 0.00; % 1 - Charging
        1.00 0.20 0.20; % 2 - Connected, not charging
        0.00 0.45 0.74; % 3 - Fully charged
    ];
    colormap(cmap);
    clim([0 3]);

    set(gca, ...
        'YDir', 'normal', ...
        'FontSize', 14, ...
        'FontWeight', 'bold', ...
        'LineWidth', 1.5, ...
        'TickDir', 'out', ...
        'Box', 'off');
    axis tight;

```

```

xticks(1:4:97);
xticklabels(0:1:24);
xlabel('Time (Hours)', 'FontSize', 13, 'FontWeight', 'bold');
ylabel('EV Index', 'FontSize', 13, 'FontWeight', 'bold');
title(sprintf('Coordinated EV Charging Schedule\nStation %d (Bus %d)', ...
    s_plot, bus_labels(s_plot)), 'FontSize', 15, 'FontWeight', 'bold');

cb = colorbar;
cb.Ticks = [0.375 1.125 1.875 2.625];
cb.TickLabels = {'Absent', 'Charging', 'Connected - not charging', 'Fully Charged'};
cb.FontSize = 11;
cb.LineWidth = 1;

% Cell gridlines
hold on;
gridColor = [0.50 0.50 0.50];
for x = 0.5:1:num_intervals+0.5
    plot([x x], [0.5 n+0.5], 'Color', gridColor, 'LineWidth', 0.2);
end
for y = 0.5:1:n+0.5
    plot([0.5 num_intervals+0.5], [y y], 'Color', gridColor, 'LineWidth', 0.2);
end

% Arrival (►) and departure (◄) markers
for i = 1:n
    plot(arrival{s_plot}(i)-1, i, 'k>', ...
        'MarkerFaceColor', 'k', 'MarkerEdgeColor', 'w', 'MarkerSize', 4);
    plot(departure{s_plot}(i), i, 'k<', ...
        'MarkerFaceColor', 'k', 'MarkerEdgeColor', 'w', 'MarkerSize', 4);
end
hold off;
set(gca, 'Layer', 'top');

% Save the figure
filename = fullfile(saveFolder3, sprintf('%d - EV Charging schedule bus %d.png', s_plot,
    bus_labels(s_plot)));
print(gcf, filename, '-dpng', '-r0');

end

```

Appendix 7 – PSO algorithm code

```
function N_opt = run_PSO(N_avail_vec, E_remaining_total, num_stations, h, p_pv, V_prev1, V_prev2, V_prev3,
t, dt, P_charger)

% ----- PROBLEM DEFINITION -----

CostFunction = @(x) evaluate_cost(x, h, p_pv, V_prev1, E_remaining_total, V_prev2, V_prev3, t, dt,
P_charger);

nVar = num_stations;
VarSize = [1 nVar];

VarMin = zeros(1, nVar);
VarMax = N_avail_vec;

% ----- PSO PARAMETERS -----

MaxIt = 40;
nPop = 20;

w = 1;
wdamp = 0.95;
c1 = 1.5;
c2 = 2.0;

VelMax = 0.2*(VarMax - VarMin);
VelMin = -VelMax;

% ----- INITIALIZATION -----

empty_particle.Position = [];
empty_particle.Cost = [];
empty_particle.Velocity = [];
empty_particle.Best.Position = [];
empty_particle.Best.Cost = [];

particle = repmat(empty_particle, nPop, 1);

GlobalBest.Cost = inf;

for i = 1:nPop
    particle(i).Position = VarMin + rand(VarSize).*(VarMax - VarMin);
    particle(i).Velocity = zeros(VarSize);
    particle(i).Cost = CostFunction(round(particle(i).Position));

    particle(i).Best.Position = particle(i).Position;
    particle(i).Best.Cost = particle(i).Cost;

    if particle(i).Best.Cost < GlobalBest.Cost
        GlobalBest = particle(i).Best;
    end
end

% ----- PSO MAIN LOOP -----

for it = 1:MaxIt
    for i = 1:nPop

        particle(i).Velocity = w*particle(i).Velocity ...
            + c1*rand(VarSize).*(particle(i).Best.Position - particle(i).Position) ...
            + c2*rand(VarSize).*(GlobalBest.Position - particle(i).Position);

        particle(i).Velocity = max(particle(i).Velocity, VelMin);
        particle(i).Velocity = min(particle(i).Velocity, VelMax);

        particle(i).Position = particle(i).Position + particle(i).Velocity;

        IsOutside = (particle(i).Position < VarMin | particle(i).Position > VarMax);
        particle(i).Velocity(IsOutside) = -particle(i).Velocity(IsOutside);

        particle(i).Position = max(particle(i).Position, VarMin);
        particle(i).Position = min(particle(i).Position, VarMax);

        particle(i).Cost = CostFunction(round(particle(i).Position));

        if particle(i).Cost < particle(i).Best.Cost
            particle(i).Best.Position = particle(i).Position;
        end
    end
end
```

```

        particle(i).Best.Cost    = particle(i).Cost;

        if particle(i).Best.Cost < GlobalBest.Cost
            GlobalBest = particle(i).Best;
        end
    end
end

w = w * wdamp;
end

% ----- OUTPUT -----

N_opt = round(GlobalBest.Position);
N_opt = max(N_opt, VarMin);
N_opt = min(N_opt, VarMax);

end

```

Appendix 8 – PSO cost function code

```
function J = evaluate_cost(N_candidate, h, p_pv, V_prev1, E_remaining_total, V_prev2, V_prev3, t, dt,
P_charger)

% ----- PARAMETERS -----

Vmin = 0.94;
Vmax = 1.06;
Trans_limit = 10; % MVA transformer rating

% Weights (tune if needed)
w1 = 1e9; % voltage violation
w2 = 1e6; % voltage fluctuation
w3 = 1e-3; % Energy backlog
w4 = 1e6; % transformer overload

% ----- LOAD FLOW -----

[Voltage, ~, S_bus, ~, ~] = Loadflow_33bus_PV_EV(h, p_pv, N_candidate);

% ----- OBJECTIVE TERMS -----

%% J1 – VOLTAGE VIOLATION (STRICT)

over = max(Voltage - Vmax, 0);
under = max(Vmin - Voltage, 0);

% J1 = sum(over.^2 + under.^2);
J1 = sum(over + under);

%% J2 – VOLTAGE FLUCTUATION AT BUS 18 (ROLLING MEAN SMOOTHING)
% Penalises deviation of current bus-18 voltage from the 3-interval
% rolling mean of the previous three voltages. This damps oscillations
% caused by PV intermittency without needing a fixed time-window gate.

V18_roll = (V_prev1(18) + V_prev2(18) + V_prev3(18)) / 3;
if t >= 30 && t <= 60
    J2 = 1000*(Voltage(18) - V18_roll)^2;
else
    J2 = 0;
end

%% J3 – ENERGY BACKLOG
J3 = E_remaining_total -(sum(N_candidate) * P_charger * dt);

%% J4 – TRANSFORMER OVERLOAD PENALTY (THERMAL STABILITY)
% Quadratic penalty above the 10 MVA thermal limit.
Trans_load = abs(S_bus(1)) / 1000; % kVA → MVA
J4 = max(Trans_load - Trans_limit, 0)^2;

% ----- TOTAL COST -----

J = w1*J1 + w2*J2 + w3*J3 + w4*J4;

end
```